Discreteness, Adjacency and Order: A Structural Comparison of the Theravāda Abhidhamma with Causal Sets and Loop Quantum Gravity

A four-row lens written for a reader who knows calculus but not physics — what each correspondence is, where it holds, and exactly where it breaks

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Abstract

This study compares the Theravāda Buddhist Abhidhamma with two discrete approaches to quantum gravity, causal set theory and loop quantum gravity, as formal structures, on three axes: order, adjacency and ratio. It is a lens and not a physics claim. No row predicts a measurement, and none says that a Buddhist text anticipated a theory of quantum gravity. Each row gives a primer of the physics, the Abhidhamma material with its textual layer named, the correspondence, and where it breaks. On order, the twenty-four conditioning relations of the *Paṭṭhāna*, taken whole, are not a partial order. The condition-modes that always run from earlier-arising to later-arising form an acyclic typed relation over which a strict order can be read: acyclic by selection, not yet a causal set, and ordering some moments of different mental streams, which share no clock. On adjacency, the strongest row, a spin-network link is the face two neighboring grains of space share and carries a quantized area, while a Theravāda sub-commentary defines space as the non-mixing of touching neighboring material clusters (*kalāpa*). Both put the unit of spatial boundary at the adjacency between minimal units, and neither makes it one of the units. The features specific to loop quantum gravity, a quantized magnitude and a volume derived from it, are the ones that fail, and the shape that holds is generic to any cellular picture. On ratio, most material phenomena last seventeen mind-moments in the commentary and the manual, and only the generic part survives: a duration stated as a count of steps. A per-kalāpa count taken as the same in every frame would conflict with muon time dilation, but the texts state no such count. Continuity, magnitude and number fall among designations for mind and matter alike on one reported method of the teachers, and the rows about mind alone constrain no matter. The one textual size for a kalāpa, read as a metric size and carried to external matter, collides with nuclear structure by count and by size; Ledi Sayadaw's reading of the same line escapes the collision and leaves matter no grain of extent. A fifth row, the conditioned as a whole against quantum causal histories, was dropped for a stated reason. Each finding carries its falsifier.

Keywords: causal set theory, loop quantum gravity, spin networks, area spectrum, quanta of space, partial order, Hasse diagram, discrete spacetime, relational space, Buddhism and physics, Buddhist philosophy of space, Abhidhamma, Paṭṭhāna, kalāpa, paricchedarūpa, ākāsadhātu, Theravāda, Abhidharmakośa, momentariness, asaññasatta, time dilation, muon lifetime, cosmic time, nuclear scattering, Rutherford scattering, proton radius, Z boson width, preorder, Lean 4, formal verification.

Terms

The paper uses Pāli technical vocabulary because it reads Pāli texts. An indexer searching in standard terms should find each concept under the term in the middle column. The layers in the right-hand column are the textual layers set out under *How to Read This Paper*.

Term as the tradition writes it	Standard term used in this paper	Layer where it is defined
S (this paper's notation)	the acyclic sub-relation formed by the condition-modes that always run from earlier-arising to later-arising (§3.3)	this paper's analysis
paccaya; *Paṭṭhāna*	condition; the canonical treatise on the 24 conditioning relations	canon (summarised in the manual, ch. 8)
anantara, *samanantara*	proximity, contiguity (immediate succession of mind-moments)	canon
āsevana	repetition	canon
natthi, *vigata*	absence, disappearance	canon
sahajāta	co-nascence (arising together)	canon
aññamañña	mutuality	canon
purejāta	pre-nascence (the condition arose earlier and still exists)	canon
pacchājāta	post-nascence (the condition arises later than what it supports)	canon
sahajāta-, *purejāta-*, *pacchājāta-* modes of *nissaya*, *indriya*, *vippayutta*, *atthi*, *avigata*	the co-nascent, pre-nascent and post-nascent modes (sub-types) of support, faculty, dissociation, presence and non-disappearance	canon (e.g. *Paṭṭhāna*: *atthipaccayena paccayo – sahajātaṃ, purejātaṃ …*); grouped in the manual, ch. 8
kamma (*nānākkhaṇika*)	volitional action (across different moments)	canon (*Paṭṭhāna*, e.g. *Kusalattika* §427: *nānākkhaṇikā – kusalā cetanā vipākānaṃ khandhānaṃ …*); the time classification is sub-commentarial
ārammaṇa	object condition	canon
upanissaya, *pakatūpanissaya*	decisive support; natural decisive support	canon
citta, *cittakkhaṇa*	a moment of consciousness; a mind-moment	canon; the durations are commentarial
uppāda, *ṭhiti*, *bhaṅga*	arising, presence, dissolution (the three sub-moments)	commentary and manual
rūpa-kalāpa	smallest material cluster	commentary; systematised in the manual
suddhaṭṭhaka	the "pure octad" (four great elements, colour, odour, taste, nutritive essence)	commentary and manual
ākāsadhātu	the space element	canon
paricchedarūpa	delimiting matter (space as the delimitation of material clusters)	manual and sub-commentary
paricchedākāsa; *kasiṇugghāṭimākāsa*; *ajaṭākāsa* (also *nijjaṭākāsa*)	delimiting space; the space left by removing a meditation disk; unentangled (open) space	commentary (*Kathāvatthu* commentary; *Atthasālinī*)
paññatti	concept, designation (as opposed to an ultimately real phenomenon)	commentary; manual
anipphanna	not concretely produced; having no intrinsic nature of its own	commentary (*Visuddhimagga*, *Sammohavinodanī*); manual
padaṭṭhāna	proximate cause	commentary
paramāṇu	the smallest unit on the commentarial size ladder; given as the kalāpa's size by one sub-commentary, and declined by Ledi Sayadaw (§4.2)	commentary; sub-commentary; modern exegesis
upanidhāpaññatti, *upanikkhittapaññatti*, *santatipaññatti*	designation by relation (long, short, far, near); by placing in series (two, three, four); by continuity ("eighty years old")	the teachers' method, reported in the commentary (§2)

Term as the tradition writes it	Standard term used in this paper	Layer where it is defined
kālavimutta	free of time (said of nibbāna and concepts)	manual; glossed in the sub-commentary
saṅkhāra, *saṅkhata*, *asaṅkhata*	the conditioned; conditioned; unconditioned	canon
sampayutta; *ekuppādanirodha*	conjoined (mind with its mental factors); one in arising and ceasing	canon; the formula is the manual's (ch. 2 §1)
asaññasatta, *asaññābhava*	non-percipient (mindless) beings; non-percipient existence, of one aggregate (matter) only	canon (DN 1; *Vibhaṅga*; *Dhātukathā*; *Kathāvatthu* III.11); commentary; manual
indriyabaddha, *anindriyabaddha*	bound to faculties (in a living body); bound to no being	commentary (*Atthasālinī*)
utusamuṭṭhāna rūpa	temperature-born matter, produced by the fire element inside and outside beings	manual (ch. 6)
kappa	aeon, world-cycle	canon (SN 15)
accharāsaṅghāta; *mattā*	a finger-snap; a mora (the time of a short vowel)	commentary; grammar
santati, *santāna*	series, continuity (of material or mental phenomena)	commentary
saṇṭhāna	shape (had by matter, lacked by mind)	commentary (*Paṭṭhāna*); sub-commentary

Findings disclosed

This paper is a study. It discloses no mechanism, so it carries findings and not claims. Each finding is stated in its surviving form, after the scope repairs of the 2026-09-28 and 2026-09-30 revisions and the additions of the 2026-11-03 revision, with its break beside it. The falsifiers are collected in §10.

1. **Order (§3.3–§3.4).** The *Paṭṭhāna*'s twenty-four conditions, taken whole, are not a partial order: mutuality makes 2-cycles, co-nascence relates what a causal order leaves unrelated, post-nascence runs from later to earlier, and the object condition reaches the future and the time-free. The condition-*modes* that always run from earlier-arising to later-arising form a relation S (proximity, contiguity, repetition, absence, disappearance, pre-nascence, kamma across moments, and the pre-nascent modes of support, faculty, dissociation, presence and non-disappearance), over which a strict order can be read. S also orders moments of different streams, with no shared clock, through kamma-born colour seen by another stream; mind-born colour gives a second route through a co-nascent edge outside S, and that route too is strict. **Break:** S is acyclic because it was selected by an index of arising time; its closure, and every cross-stream composition, is the reader's; local finiteness is unexamined; and S alone conditions no arising. So S is not a causal set.
2. **Adjacency (§4.5, narrowed by §9).** Loop quantum gravity and the Theravāda sub-commentary both put the unit of spatial boundary at the adjacency between minimal units, and neither makes that boundary one of the units. On *what is between minimal units?*, the *Vibhāvinī*'s answer (a named, counted item of the material list, though not an ultimate reality) has the shape of loop quantum gravity's (a link), not the Kashmiri Sarvāstivāda's (*nothing*). The tradition gives its own reason why the between is matter's alone: mind has no shape. **Break:** the parts of the match specific to loop quantum gravity (a quantised magnitude on the boundary, volume derived from it) are the parts that fail. The dependence runs in opposite directions; the link is incident to both grains, the delimiter touched by neither; the delimiter has no magnitude and is not an ultimate reality; kalāpas are matter in space, nodes quanta of space; the canonical space element is a cavity. The shape that holds is also held by any cellular picture and by the "neither" view of boundaries (§9), and the matter–mind asymmetry is the chain-against-antichain shape.
3. **Ratio (§5.3, narrowed by §9).** Material phenomena other than the two intimations and the characteristics last seventeen mind-moments (fifty-one sub-moments) in the commentary and the manual; the tradition also records sixteen and a third, and sixteen (§5.2). Where a stream is present, the commentary dates its mind-born and kamma-born matter on its own mind-moments without making mind that matter's condition. Where none is present, no text counts matter in mind-moments: the mindless beings' material series is timed by kamma's momentum in aeons, and a sub-commentary makes the mind-moment a unit. What survives of the comparison is generic: a duration is a count of steps. **Break:** every discrete clock shares that much; the longest chain and the statistics fail, and whether a stream's order makes its time is left open (§3.4). A per-kalāpa count taken as the same in every frame would conflict with muon time dilation, but the texts state none, and let a series' duration vary (§5.4). The aeon meets the mind-moment only as divergence.
4. **The dropped row (§6).** *Saṅkhāra* (the conditioned as a whole) set against quantum causal histories adds no structure beyond findings 1–3, and it maps a category that contains mind onto a structure that contains none. **Break:** the row would return if a structural feature of quantum causal histories were shown to have a *Paṭṭhāna* counterpart not already in findings 1–3.

5. **The scale of the mind-moment (§5.2).** The texts name the finger-snap, the unit of their rate of mind, by a gesture and, in a grammar, as a mora or an eye-blink, without giving it a duration. They state the rate as round lower bounds in four works, of 10^{12}, 10^{10} and 10^{7} mind-moments per snap or more, and as a heap of rice grains; none states a ceiling. **Break:** any figure in seconds is the reader's conversion. The texts deny discernment of the succession to the reasoner and to "others", not to the wise or the Buddha. A text giving the snap a duration, or the rate an upper bound, would change this finding.
6. **The status of continuity (§2).** On the teachers' method reported in the *Puggalapaññatti* commentary, relative length and distance, number, shape and continuity-age are designations on the unreal side, for mind and matter alike. The rows of the formal model that concern mind alone constrain no matter; matter's temporal discreteness enters only through the rows that state it. *Santati*, where each mind-moment has an immediate successor, is the gaplessness of a discrete series and not a continuum. So the texts settle the status of continuum-talk, and not the structure of matter. **Break:** the method is reported as the teachers', not the canon's; individuated dhammas stay real; the no-constraint result holds by the model's signature; the density argument covers the mental stream only, since a sub-commentary denies proximity to matter.
7. **The grain's count and size (§4.2, §4.8).** The tradition has its own bridge from time to count to size: a sub-commentary on the *Abhidhammāvatāra* gives forty-nine kalāpas from a seventeen-moment life and three sub-moments, beside a measure bounded by the *paramāṇu*. The count follows from time, and the size does not. The *Visuddhimagga Mahāṭīkā*'s "those kalāpas are of *paramāṇu* size", read as the metric size of one kalāpa and carried to external matter, collides with nuclear structure by count and by size on every anchor tried. Ledi Sayadaw reads the line as conformity with the powder meditation; his bound gives one kalāpa no lower size, and no grain of extent for matter follows. **Break:** the anchor, the carrying outward and the nucleon's place in the count are this paper's; every textual size concerns bodily matter; the line is a purpose clause in a meditation instruction, and a second reading in Ledi takes it of an accumulated measure.
8. **A conditional bound on the sub-moment (§5.4).** If the Z boson were a series (*santati*) of pure octads, which is this paper's supposition, its measured width would put one sub-moment at or below about 2.6×10^{-25} seconds. **Break:** the supposition is in no text; it gives an upper bound only, with no floor; a stronger bound of fifty-one times, tried in the source run, does not follow from the same premises.

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What is already public, and cited rather than claimed:

- **Causal-set becoming linked to a Buddhist school.** Rafael Sorkin, a founder of causal set theory, remarked in 2007 that the "accretive time" of his growth models "seems close to that … of the 'Vibhajavadin' school within the Buddhist philosophical tradition". Fay Dowker quoted the remark in 2022 and invited a scholar to examine the earliest records. This institution's sibling paper, *the-vibhajjavadin-view-of-time*, answers that invitation: the Pāli Vibhajjavāda is presentist, and so is the **foil** of causal-set growth, not its twin. This paper does not revisit that finding and does not contradict it.
- **Causal sets as the discrete theory of order.** Bombelli, Lee, Meyer and Sorkin (1987); Rideout and Sorkin (2000) on sequential growth; Wüthrich and Callender (2017) on what kind of becoming a causal set allows.
- **Quanta of space and the discrete area spectrum.** Rovelli and Smolin (1995); Ashtekar and Lewandowski (1997); Rovelli and Upadhya's primer (1998); Rovelli's Zakopane lectures (2011); Bianchi, Doná and Speziale (2011) on spin-network nodes as polyhedra.
- **Syntheses of order and spin networks.** Markopoulou and Smolin (1997); Markopoulou (2000), *quantum causal histories*; Hawkins, Markopoulou and Sahlmann (2003).
- **Time dilation of moving lifetimes, measured.** Rossi and Hall (1941), on the decay of cosmic-ray mesotrons (muons) by momentum; Bailey et al. (1977), on muons circulating in the CERN storage ring. §5.4 uses both against one reading of the seventeen-moment lifetime.

- **This institution's own earlier work, cited rather than repeated.** The paper *Twenty-Four Kinds of Because* (§4.7) finds that no *Paṭṭhāna* condition joins two mental continua and that two people have no shared clock; §3.4 below narrows that from the other side, since no shared clock is not no order. The Lean 4 repository `SiliconWat/formal-abhidhamma` (CC0; software concept DOI 10.5281/zenodo.23067241) holds the machine check cited in §4.5, the run that supplied three of §5.2's four lower bounds, and the run *the continuum bridge* (`runs/2026-10-14-the-continuum-bridge/`, commit `a964ee1`), the source of §2's added finding, §4.8 and §5.4's bound.
- **Why the between is matter's and not mind's.** The *Paṭṭhāna* commentary and sub-commentaries give the reason (mind has no shape), and Ledi Sayadaw's *Paṭṭhānuddesa-dīpanī* (c. 1900) states the contrast outright (§4.5). This paper claims nothing in it.
- **Buddhism read against physics in general.** Carlo Rovelli's *Helgoland* (2021) reads relational quantum mechanics beside Nāgārjuna. Popular and scholarly comparisons of Abhidhamma momentariness with quantum physics exist, among them *The Ephemeral Projector* (ResearchGate). An earlier scholarly instance carries in its title the anticipation reading this paper declines: Shanta Ratnayaka, "Quantum physics foreshadowed in the Abhidhamma" (*Sri Lanka Journal of Buddhist Studies* 3, 1991), cited by its bibliographic record only; it was not read for this paper. A 2026-07-31 blog post, self-labeled AI-generated and without citations, compares causal set theory and the Abhidhamma in general terms (ekasiswanto.wordpress.com). The Grokipedia article "Kalapa (atomism)" compares the kalāpa's size to the Planck length. The Wikipedia article of the same name, read on the date of writing, does not.
- **The scholarly expositions of the Abhidhamma material this paper reads.** None of these compares the material with physics. Each already expounds a textual point the rows depend on.
- Y. Karunadasa, *Buddhist Analysis of Matter* (1967), chs. 5 and 7. It reads the space element as cavities and interstices, "bounded or delimited space", with *pariccheda* meaning both what delimits and what is delimited. It sets the Theravāda space element beside the Vaibhāṣika pair (unconditioned *ākāśa*; the conditioned *ākāśadhātu* of door, window, mouth and nose). It reports the *Kathāvatthu*'s two-trees argument. And it explains pre-nascence and post-nascence by the longer duration of matter than of mind. **It cuts against the adjacency row in one respect.** On its reading, the later scholiasts' classing of the space element as *anipphanna* means that it is "neither *rūpa* … nor a *dhamma* … in the true sense", "nothing but the mere limitation of matter". §4.4 and §4.6 carry this.
- Bhikkhu Bodhi (gen. ed.), *A Comprehensive Manual of Abhidhamma* (1993), ch. VI and ch. VIII. It sets out the sub-types (modes) of each condition, among them base-prenascence support, prenascence faculty (the five sensitivities, "arisen at the static phase of the past bhavaṅga citta"), and the prenascent and postnascent types of dissociation, presence and non-disappearance. It also states that non-concretely produced matter, the space element included, is "not included among the ultimate realities". Its introduction dates the seventeen-moment lifetime of matter, the three sub-moments and the kalāpa to the commentaries.
- U Nārada Mūlapaṭṭhāna Sayadaw (with Thein Nyun), *Conditional Relations* (*Paṭṭhāna*), vol. I (PTS Translation Series 37, 1969). An English translation of the Chaṭṭha Saṅgāyana *Paṭṭhāna* with an introduction and charts that work through each condition's modes (for example, "base-prenascence-dependence"). It is the route for the canonical-layer check of row 1 that §11 lists as owed.
- Alexander von Rospatt, *The Buddhist Doctrine of Momentariness: A Survey of the Origins and Early Phase of this Doctrine up to Vasubandhu* (1995). The standard study of the doctrine of moments. **Cited by its bibliographic record only: it was not reached for this revision**, so the paper does not characterise what it says about the seventeen-moment lifetime.
- **The Indian debate on whether minimal material units touch** (*Abhidharmakośabhāṣya* ad I.43, cross-tradition, §4.4 below). It is the nearest Indian neighbor of the adjacency row that this paper reads, and it predates every physics comparison by more than a millennium. Older Western accounts of boundary and place, Aristotle's among them, are cited in §9.

The census. A quick novelty census ran on 2026-09-27. Its pre-registration was pushed publicly before any query was made: `thonly/publications` commit `7348ba2`, `timestamps/census/physics-lens-prereg.md`. The search covered the English web (including dharma sites and popular Buddhism-and-physics writing), and arXiv, Google Scholar, PhilPapers, GitHub and Google Patents by keyword. Thai, Burmese, Sinhala, Chinese, Japanese and Khmer sources were **not run**, and no books were read. The pre-registered control, the known comparison *The Ephemeral Projector*, was found.

Conjunct	Found	Verdict
(a) the *Paṭṭhāna*'s typed conditions against causal-set order	a general causal-set–Abhidhamma resonance (the AI-generated post; names no *Paṭṭhāna* condition)	**narrowed** at general width; the typed-relation comparison was not found
(b) kalāpa against spin-network nodes	kalāpas compared with atoms, quanta and the Planck length in popular writing	**narrowed** at "smallest unit" width; the spin-network-specific comparison was not found
(c) *paricchedarūpa* (space as the between) against area on spin-network links	the commentarial reading of space as delimitation is public; no comparison with loop quantum gravity	**not found in that aperture on that date**
(d) *saṅkhāra* against quantum causal histories	—	not found; **dropped before drafting** (§6)
(e) the 17-moment lifetime as a discrete-time ratio	Sorkin and Dowker link causal-set becoming to Buddhist momentariness	**narrowed** at general width; the chain-length reading of the ratio was not found
the composition of rows in one structural map	—	**not found in that aperture on that date**

"Not found in that aperture on that date" is a statement about a search. It is not a claim of priority, and nothing in this paper is offered as new. As a study that makes no priority claim, this paper owes no further census before deposit (ruled 2026-09-28); a literature census could narrow or kill the rows above, and would be reported here as a revision.

> **Note.** As first published on 2026-09-27 this paper carried the CC0 dedication and the statement that its authors and HeartBank® would not seek patent on anything in it; it carried no commitment not to assert. The commitment not to assert, and the extension of both commitments to the other entities named above, were added, and the statement brought to the standard form above, on 2026-10-06. The omission did not affect the paper's standing as prior art, which publication establishes; the date relied on is the one carried by this document's OpenTimestamps proof.

How to Read This Paper

This paper is written so that a reader who knows calculus, but no linear algebra, quantum mechanics or general relativity, can follow every step. Each physics section opens with a subsection marked **Physics primer**, which builds the idea from nothing. A reader who already knows the physics can skip those subsections. Everything else is written for the exhaustive reader: the Pāli is quoted, the textual layer of every Abhidhamma claim is named in words, and every correspondence ends with the place where it fails.

Formulas are written in plain Unicode. Display formulas sit in monospace blocks. Where a formula is quoted, the source that derives it is cited.

Four textual layers are kept apart throughout, with a grammar and a modern exegete used for single points. They are named in words at each use, and every Pāli passage added in the 2026-09-30, 2026-10-06 and 2026-11-03 revisions is cited by its Chaṭṭha Saṅgāyana file and line (for example `abh07t.nrf.xml:1485`) in the VRI `tipitaka-xml` text at commit `05d5d3c`:

Layer	Texts used here	Date (traditional or scholarly)
canon	*Dhammasaṅgaṇī*, *Vibhaṅga*, *Dhātukathā*, *Kathāvatthu*, *Paṭṭhāna* (Abhidhamma Piṭaka); for §5 also the *Brahmajāla Sutta* (DN 1), the *Saṃyutta*'s chapter on the beginningless round (SN 15), and the *Milindapañha* (also §7), canonical in this edition only	compiled over the last centuries BCE; the *Kathāvatthu* is traditionally dated to the Third Council
commentary	*Visuddhimagga*, *Sammohavinodanī* (*Vibhaṅga* commentary), *Atthasālinī* (*Dhammasaṅgaṇī* commentary), the *Kathāvatthu*, *Puggalapaññatti* and *Paṭṭhāna* commentaries; for §5 also the commentaries on the *Dīgha*, *Saṃyutta*, *Aṅguttara* and *Suttanipāta*	c. 5th century CE
manual	*Abhidhammatthasaṅgaha* (Anuruddha)	variously dated; commonly c. 11th–12th century
sub-commentary	*Abhidhammatthavibhāvinī-ṭīkā* (Sumaṅgala); also the *Mūlaṭīkā* on the *Vibhaṅga* and on the *Paṭṭhāna* with their *Anuṭīkā**s, the **Mūlaṭīkā* on the *Dhammasaṅgaṇī* (§7 only), the *Dhammasaṅgaṇī Anuṭīkā*, the *Dīgha* and *Aṅguttara* sub-commentaries, the *Sāratthadīpanī* (on the Vinaya), the *Visuddhimagga Mahāṭīkā* and a sub-commentary on the *Abhidhammāvatāra* (§4.2 and §4.8 only)	c. 12th century for the *Vibhāvinī*; the *Mūlaṭīkā* is earlier, since the *Vibhāvinī* names its author (*Mūlaṭīkākārādayo*, `abh07t.nrf.xml:5553`)
grammar	*Payogasiddhi*, a Pāli grammar, used in §5.2 for one unit only	medieval
modern exegesis	Ledi Sayadaw, *Paramatthadīpanī* (on the manual), its *Anudīpanī* (his own, by its colophon) and *Paṭṭhānuddesa-dīpanī*	c. 1900

A claim that holds only at a later layer is not a claim about the canon. Several rows below live, or die, on exactly that distinction.

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1 · What Kind of Comparison This Is

Physics has two great theories that do not fit together. General relativity describes gravity as the smooth geometry of spacetime. Quantum theory describes everything else as discrete, probabilistic excitations. A theory of **quantum gravity** would describe what spacetime is at the smallest scale, where both matter. Two of the candidate programmes make spacetime itself discrete:

- **Causal set theory** says spacetime, at bottom, is a set of events and a relation of *before*. Everything else, including distance and dimension, is to be recovered from counting.
- **Loop quantum gravity** says space, at bottom, is a network of grains whose areas and volumes come in discrete steps.

The Theravāda Abhidhamma is also a discrete analysis. It resolves experience and matter into momentary phenomena (*dhamma*), connected by twenty-four typed conditioning relations. It gives matter a smallest cluster, and it gives space a definition.

It is therefore tempting to line the two up. That temptation has a long and mostly poor history, from *The Tao of Physics* onward, and this paper is built to resist it. Every row below is a **structural** comparison: it asks whether two frameworks share a *shape* (an order, an adjacency, a ratio), and it names exactly where the shape stops being shared. The Abhidhamma is an analysis of what can be known in experience. Quantum gravity is a theory of what spacetime is. Whether the two can be compared at all is the strongest objection to this paper, and §8 carries it rather than answering it.

The map, in one table:

ROW	PHYSICS SIDE	ABHIDHAMMA SIDE	SURVIVING FORM
1 ORDER	causal set: a locally finite partial order ("before")	the Paṭṭhāna's 24 typed conditions (classified by manual + sub-comm.)	only the earlier→later MODES (S) are acyclic, by selection: a strict order, not yet a causal set; S also orders across streams (kamma-born colour); the rest is typing a causal set lacks
2 GRAIN	spin-network node: a quantum OF space, with a volume	rūpa-kalāpa: smallest material cluster (matter IN space)	survives only as what row 3 presupposes; matter ≠ geometry; its one textual size collides with nuclear structure (§4.8)
3 BOUNDARY	spin-network link: the face two grains share; carries quantised area	paricchedarūpa: the non-mixing of touching neighbours; "limit of matter"	both put the unit of spatial boundary at the ADJACENCY, not in a unit (strongest row; the shared part is generic; seven breaks)
4 RATIO	proper time ∝ length of the longest chain between two events	most matter lasts 17 mind-moments (51 sub-moments)	dated on a stream's own moments where there is one, a UNIT where there is none; only the generic part survives: a duration as a count
— DROPPED —	quantum causal histories	saṅkhāra (the conditioned as a whole)	category error; added no structure

Rows 2 and 3 are one argument told from two ends, so they share one section (§4).

---

2 · A Note on Discreteness

Two different things are called *discrete* in this paper, and they must not be confused.

- **Object discreteness.** The world is made of separate pieces, like grains of sand. The kalāpa is discrete in this sense, and so are the elements of a causal set.
- **Spectral discreteness.** A *measured quantity* can take only values from a list with gaps, like the rungs of a ladder rather than a ramp. Loop quantum gravity is discrete in this sense. Rovelli warns explicitly (Zakopane lectures, §I.A) that its discreteness "is not given by the fact that the grains of space … are discrete objects. Rather, it is given by the fact that the size of each grain is quantized in discrete steps."

The Abhidhamma has object discreteness in abundance: moments, clusters, sub-moments. Its quantities are restricted only as **counts** of discrete units: a duration is a whole number of sub-moments, and a size is a step on a ladder of powers of thirty-six (§4.2). It never quantises a magnitude that is not a count, and that is what spectral discreteness is. In particular, its delimiting space (manual and sub-commentary) is assigned no magnitude at all (§4.6). This single fact limits every row below.

Continuity and magnitude are designations, for mind and for matter. The *Puggalapaññatti* commentary reports a method "outside the commentary, of the teachers" (*aṭṭhakathāmuttakena pana ācariyanayena*, `abh03a.att.xml:741`). It places "long, short, far, near" among designations by relation (*upanidhāpaññatti*, `:745`), "two, three, four" among designations by placing in series (*upanikkhittapaññatti*, `:753`), and "eighty years old", a designation that looks to the absence of a break in continuity (*santativicchedābhāvaṃ apekkhitvā*), among designations by continuity (*santatipaññatti*, `:753`). Of that list only the designation of a dhamma's own nature (earth, heat, hardness) is on the real side; the rest "belong to the unreal side" (*avijjamānapakkhañceva*, `:753`). Time and direction are designations depending on the turning of sun and moon (`:741`), and shape designations are also unreal (`:757`). Three limits apply. This is the teachers' method as reported, not the canon. It makes the numeral unreal and leaves the plurality of dhammas real. And it does not make all designation unreal: some place-designations and the designation of the unconditioned name what exists (`:757`).

This settles a status and not a structure. The rows of the formal model that concern mind alone (`kala/AXIOMS.md`, A1, A3, A7, A24, A104, A112, A117, A126–A128) contain no symbol for matter, so they constrain none; that is true by construction (`d3_inner_rows_say_nothing_of_matter`). Matter's temporal discreteness enters only through the rows that state it: the seventeen-moment life (A4), the *Visuddhimagga*'s clause on matter's sub-moments in A40, production (A64) and the kalāpas (A66). So the discreteness of the mental stream entails no discrete outer base.

Nor is the texts' *continuity* a physicist's *continuum*. A stream in which each moment has an immediate successor cannot be dense, since between a moment and its successor no moment lies (`d2_proximity_forbids_density`, with a model showing the premises are consistent). The texts' language fits: the compactness of continuity is the going-on of earlier and later "as if become one, through their having no interval" (*purimapacchimānaṃ nirantaratāya ekībhūtānamiva pavatti santatighanatā*, `abh01t.tik.xml:681`). This holds for the mental stream only. The *Vibhaṅga Anuṭīkā* denies that relation to matter (*na hi rūpadhammānaṃ arūpadhammānaṃ viya anantarapaccayabhāvo atthi*, `abh02t.tik.xml:4921`), and describes matter's series as linked like a chain (*saṅkhalikassa viya sambandho*, `:4921`), which a dense family could also satisfy.

---

3 · Row 1 — Order

3.1 · Physics primer: partial orders and causal sets

A relation is a rule that says, for any two things x and y, whether x stands in it to y. "Is an ancestor of" is a relation on people. Write it $x \prec y$.

A partial order is a relation with two properties:

```
irreflexive:   never  x < x                      (nothing precedes itself)
transitive:    if  x < y  and  y < z,  then  x < z  (ancestors of ancestors are ancestors)
```

Together these make it **acyclic**: you can never follow the arrows and return to where you started. It is called *partial* because some pairs are not related either way. Two cousins are neither ancestor of the other.

A causal set is a partial order on a set of events, with one more property, **local finiteness**: between any two related events, only finitely many events lie. The relation means "can causally influence". In relativity, two events are either causally related (one is in the other's past light cone) or **spacelike**: too far apart in space, for the time between them, for any signal to connect them. In a causal set, spacelike simply means *unrelated*.

Links and the Hasse diagram. If $x \prec y$ and nothing lies between them, the pair is a **link** (mathematicians say *covering relation*). Drawing only the links, with later events higher, gives the **Hasse diagram**. Every other relation can be recovered by following links upward, since the order is transitive.

```
          e       f            later events higher
           \     /
            \   /
              d                only LINKS are drawn;
             /   \             a < d, a < e, a < f, b < e … follow by transitivity
            b     c
             \   /             b and c are UNRELATED (spacelike):
               a               neither can influence the other
```

Order plus number. Sorkin's slogan for the programme is *order + number = geometry*. Two classical theorems (Hawking, King and McCarthy 1976; Malament 1977) show that, for spacetimes that are past- and future-distinguishing, the causal order fixes the geometry up to a conformal factor, that is, up to a local scale. Causal set theory supplies the missing scale by **counting**. Events are scattered at random (a

"sprinkling") with, on average, one event per unit of spacetime volume in Planck units, so the number of events in a region measures its volume.

Time as chain length. A **chain** is a sequence $x^*_1 \prec {}^*x^*_2 \prec \dots \prec {}^*x^*_n$. *Between two related events, the length of the* ***longest*** *chain grows in proportion to the proper time between them: the time a clock carried from one to the other would read. The result holds asymptotically and in expectation, for random sprinklings into Minkowski space (Brightwell and Gregory 1991). In a causal set, a duration *is* a count of steps. §5 uses exactly this.

Growth. In Rideout and Sorkin's sequential-growth models, the causal set is not given all at once. It grows, one element at a time, each new element born to the future of some existing ones. What is born stays. Sorkin reads this as a real becoming, a *growing block*: a past that accumulates and a future that does not yet exist.

3.2 · The Abhidhamma side: twenty-four typed relations

The *Paṭṭhāna*, the last book of the Abhidhamma Piṭaka (canon), analyses how phenomena condition one another under twenty-four named conditions (*paccaya*). The manual (*Abhidhammatthasaṅgaha*, ch. 8) summarises them, and the sub-commentary (the *Vibhāvinī*) classifies each by the **time** its conditioning reaches. On the manual's ch. 8 it states:

> … *anantarasamanantaraāsevananatthivigatavasena pañcannaṃ atītakālikānaṃ, kammapaccayassa atītavattamānavasena dvikālikassa, ārammaṇaadhipatiupanissayapaccayānaṃ tikālikānaṃ, itaresaṃ pannarasannaṃ paccuppannakālikānañca vasena kālattayavanto, nibbānapaññattivasena kālavimuttā ca* …
>
> … [things are] possessed of the three times by way of the five belonging to the past (proximity, contiguity, repetition, absence, disappearance), the kamma condition belonging to two times (past and present), the object, predominance and decisive-support conditions belonging to three times, and the remaining fifteen belonging to the present; **and free of time, by way of nibbāna and concepts** …

The count checks: 5 + 1 + 3 + 15 = 24. The tetrad sorts the conditions by the time their conditioning state may have, and the same sentence then sets the time-free (nibbāna and concepts) beside the three times. So the classification is not exhaustive of what can condition: the object condition, placed among the three-time conditions, also reaches the time-free.

The remaining fifteen are root (*hetu*), co-nascence, mutuality, support (*nissaya*), pre-nascence, post-nascence, result (*vipāka*), nutriment (*āhāra*), faculty (*indriya*), jhāna, path (*magga*), association (*sampayutta*), dissociation (*vippayutta*), presence (*atthi*) and non-disappearance (*avigata*). "Belonging to the present" means the conditioning state exists when the conditioned one does. It does not mean that the two arose together. Pre-nascence belongs to the present because its condition arose earlier and still stands. So "present" is not "co-nascent", and several of the fifteen have **modes**, which the canon itself names. The *Paṭṭhāna* gives the presence condition as *sahajātaṃ, purejātaṃ, pacchājātaṃ, āhāraṃ, indriyaṃ* (co-nascent, pre-nascent, post-nascent, nutriment, faculty), and the dissociation condition as *sahajātaṃ, purejātaṃ, pacchājātaṃ*. The manual groups the modes (ch. 8, CST §30–§34):

Condition	Co-nascent mode	Pre-nascent mode (condition arose earlier)	Post-nascent mode
support (*nissaya*)	mind with mind, the great elements with derived matter	the six bases for the seven consciousness elements, during the course of existence	—
faculty (*indriya*)	the immaterial faculties for co-nascent mind and matter	the five sensitivities for the five kinds of sense consciousness	— (the material life faculty, for the matter of its own cluster, is a third, separate mode)
dissociation (*vippayutta*)	mind for co-nascent matter; the base at rebirth-linking	the six bases, during the course of existence, *purejātavasena*	later mind for the earlier-arisen body
presence, non-disappearance (*atthi*, *avigata*)	*sahajātaṃ*	*purejātaṃ*	*pacchājātaṃ* (edible food and material life are two further, separate modes)

The manual names the pre-nascent mode in words for dissociation, presence and non-disappearance. For support and faculty the pre-nascent reading is the one the canon gives for the base (*vatthu kusalānaṃ khandhānaṃ nissayapaccayena paccayo*) and the one the standard modern exposition gives for the sensitivities (Bodhi 1993, ch. VIII: "prenascence faculty"). At rebirth-linking the base arises together with the mind and is co-nascent, which is why the manual limits the pre-nascent mode to the course of existence.

A condition in the Abhidhamma is **typed**. The relation is not just "*x* conditions *y*", but "*x* conditions *y* **as** object", "**as** kamma", "**as** co-nascent", and so on. One pair can stand in several types at once.

3.3 · The correspondence, and its surviving form

The whole is not a partial order. Four features break it:

Feature	Text (layer)	Why it is not a partial order
Mutuality (*aññamaññā*)	canon; manual ch. 8	*x* conditions *y* and *y* conditions *x*: a 2-cycle. A partial order forbids cycles.
Co-nascence (*sahajāta*)	canon; manual ch. 8	relates what arises **together**. A causal order leaves simultaneous, unlinked events *unrelated* (spacelike).
Post-nascence (*pacchājāta*)	manual ch. 8: "*pacchājātā cittacetasikā dhammā purejātassa imassa kāyassa pacchājātavasena*" — "later-arisen consciousness and its concomitants [condition] this body, arisen earlier, by way of post-nascence"	the condition arises **after** what it supports: ordered by arising, the arrow runs backward.
Object (*ārammaṇa*)	manual ch. 3: mind-door objects may be "*paccuppannamatītaṃ anāgataṃ kālavimuttañca*" (present, past, future, and free of time); the sub-commentary glosses the time-free as "*nibbānaṃ, paññatti ca*" (nibbāna and concepts)	a future object conditions a present mind, and a time-free object is in no order at all.

The surviving form. Because several conditions have modes, the surviving relation is defined over condition-**modes**, not over whole conditions. Some modes always run from what arose **earlier** to what arises **later**:

```
the ACYCLIC SUB-RELATION S  (defined over MODES)
─────────────────────────────────────────────────────────────────────────
anantara, samanantara     the moment just ceased → the next moment
āsevana                   a moment → the next, as repetition
natthi, vigata            the moment just ceased → the next (by its absence)
purejāta                  an earlier-arisen, still-present base or object → a later-arising mind
kamma (nānākkhaṇika)      a past volition → its later result, across many moments
pre-nascent MODES of      the same earlier-arisen bases (and, for faculty, the five
  nissaya, indriya,         sensitivities) → the later-arising consciousness, during the
  vippayutta, atthi,        course of existence
  avigata
```

The co-nascent and post-nascent modes of those five conditions stay out of S, and so does kamma's co-nascent mode.

Every one of the twenty-four, assigned:

Where it goes	Conditions (or modes)
in S (whole condition)	proximity, contiguity, repetition, absence, disappearance, pre-nascence
in S (one mode only)	kamma (across moments); support, faculty, dissociation, presence, non-disappearance (pre-nascent modes)
out: co-nascent only	root, co-nascence, mutuality, result, jhāna, path, association; the co-nascent modes of kamma, support, faculty, dissociation, presence, non-disappearance, predominance and nutriment
out: runs backward	post-nascence; the post-nascent modes of dissociation, presence and non-disappearance
out: reaches three times, or the time-free	object, predominance (object mode), decisive support
out: not sorted by arising here	nutriment (edible food); faculty and presence (material life), whose relata co-exist in one body without a stated arising order

Why S is acyclic (a one-line proof). Let $t(x)$ be the moment at which phenomenon x arises. For every mode in S, x S y implies $t(x) < t(y)$. Along any chain x^*_1 S x^*_2 S … S x^*_n, *the arising times strictly increase, so the chain can never return to* x^*_1. ***This proof shows less than it seems to.*** *S was *selected* by the condition $t(x) < t(y)$, so its acyclicity follows from the selection and is not a finding about the text. A strict order can be read over S, and its transitive closure is a partial order. But the closure is the reader's construction (see the next paragraph), and whether S meets the rest of §3.1's definition is taken up in §3.4.

One condition was removed from S by the adversarial review. The original list included *pakatūpanissaya* (natural decisive support). The sub-commentary classes decisive support as **tikālika**, reaching all three times (quoted above), and the manual's list of natural decisive supports includes present things (a person, climate, food, lodging). It is therefore not guaranteed to run earlier-to-later, and it is out.

A second, smaller correspondence: generators, not closure. The manual and the sub-commentary present the conditions as primitive. Neither says that because x conditions y and y conditions z, x conditions z. Whether the *Paṭṭhāna* itself anywhere composes two conditions

into a third is part of the canonical check this paper owes (§11); U Nārada's translation (1969), which works through the *Paṭṭhāna*'s own permutations, is the route for it. On the manual's presentation, the relations are **primitive**, like a causal set presented by its links rather than by its full transitive order. The likeness is partial. Proximity (*anantara*) holds only between immediate successors, so it is link-like in the strict Hasse sense. Kamma across moments, however, spans many intermediate moments and is still primitive, so S is a set of generating relations that is neither the Hasse diagram nor the full order.

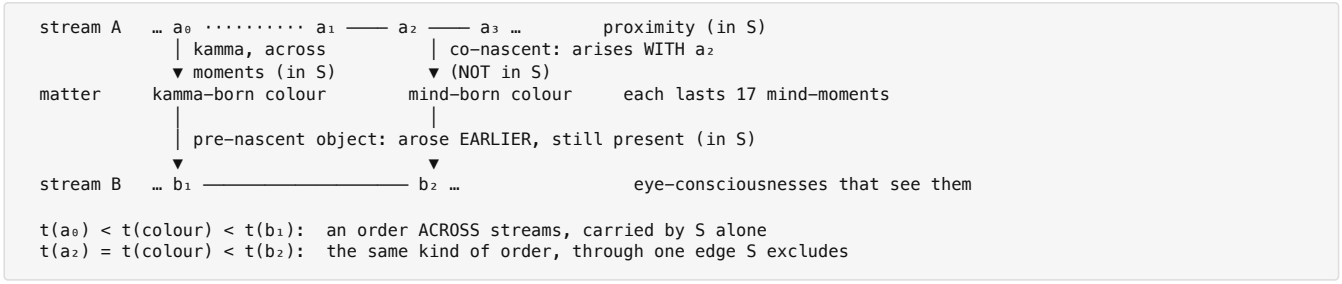
The surviving form of the row, stated once:

> **As the manual and the *Vibhāvinī* classify the *Paṭṭhāna*'s conditions, the condition-modes that always run from earlier-arising to later-arising (proximity, contiguity, repetition, absence, disappearance, pre-nascence, kamma across moments, and the pre-nascent modes of support, faculty, dissociation, presence and non-disappearance) form an acyclic typed relation over which a strict order, but not yet a causal set in §3.1's sense, can be read. The remaining conditions and modes (the co-nascent ones, the post-nascent ones, and the object, predominance and decisive-support conditions that reach all three times) are typed structure that a causal set does not have.**

3.4 · Where it breaks

- **Different kinds of order.** Causal sets keep their past: in sequential growth, what is born stays. The Theravāda is **presentist**: only present phenomena exist (the sibling paper, from *Kathāvatthu* §282–§304). S is an order of *conditioning*, not an order of *co-existing* elements. Its earlier relatum has usually ceased by the time the later arises. The sibling paper's phrase is **influence without existence**. So the two orders are not the same kind of thing, and this paper does not say they are.
- **S is selected by an index the texts do not define.** The proof above uses $t(x)$, an index of arising moments read off the texts' *earlier-arisen* and *later-arising*, and S is acyclic **because** it was selected by it. The texts make time a designation on phenomena, "a mere concept" (*paññattimattako*, *Atthasālinī*, `abh01a.att.xml:3721`), used "only on the occurrence of phenomena" (*dhammānaṃ pavattimeva ca upādāya kālavohāro*, *Paṭṭhāna Mūlaṭīkā*, `abh03t.tik.xml:4309`), and define no index across streams. In a causal set there is no time apart from the order. Whether a stream's succession stands to its time in that way, or presupposes one, this paper leaves open; its proof needs t either way.
- **S is not a causal set in §3.1's own sense.** Four gaps remain between S and that definition. (1) Acyclicity follows from the selection by t, as noted under the proof. (2) The transitive closure, which a partial order needs, adds exactly the composed relations the texts do not assert, so the closure is the reader's construction. (3) Local finiteness is unexamined: nothing here counts how many phenomena lie between two related ones, and the answer depends on the index t again. (4) S alone conditions no arising. Every arising also needs co-nascent conditions, which are outside S, so S is not a sufficient account of how anything comes to be. "A strict order can be read over S" is the whole of what survives.
- **Typing has no counterpart.** A causal set has one relation. The *Paṭṭhāna* has twenty-four, and its analytical power lies in the types. Loop-gravity syntheses add structure (spins, Hilbert spaces) to causal relations, but none of it corresponds to *object*, *predominance* or *kamma*. §6 returns to this.
- **Mind is inside the relation.** Most *Paṭṭhāna* conditions have mental relata. Nothing in a causal set is mental.

Order between streams. The sibling paper *Twenty-Four Kinds of Because* (§4.7) finds that no condition joins two mental continua and that two people have no shared clock. No shared clock is not no order, and S itself carries some. Kamma across moments runs from a past volition to its results, and the canon counts matter among them (*nānākkhaṇikā – kusalā cetanā vipākānaṃ khandhānaṃ kaṭattā ca rūpānaṃ kammapaccayena paccayo*, *Paṭṭhāna*, `abh03m7.mul.xml:6065`; manual ch. 8 §17, `abh07t.nrf.xml:2865`). Kamma-produced matter includes visible form (*rūpaṃ kammassa katattā rūpāyatanaṃ*, *Dhammasaṅgaṇī* §652, `abh01m.mul.xml:5549`), and every kamma-born cluster carries the inseparable eight, colour (*vaṇṇa*) among them (manual ch. 6 §46, §27, `abh07t.nrf.xml:2273`, `:2169`). Colour is a pre-nascent object for the five-door process of whoever sees it (*pañcārammaṇāni ca pañcaviññāṇavīthiyā purejātavasena*, manual ch. 8 §19, `abh07t.nrf.xml:2873`). Both edges are in S. A second route runs through mind-born matter: a mind originates matter by co-nascence (*cittacetasikā dhammā aññamaññaṃ sahajātarūpānañca*, ch. 8 §28, `abh07t.nrf.xml:2909`), at its own arising sub-moment (*Sammohavinodanī*, `abh02a.att.xml:409`), and mind-born octads carry colour too (ch. 6 §47, `abh07t.nrf.xml:2277`).



The carrier is the colour, never the bodily intimation, which lasts one mind-moment (§5.2) and is not visible. Three limits apply. (1) Each route is the reader's composition of two edges the texts state separately. (2) The mind-born route lives only in the larger relation of S with the co-nascent modes, whose reflexive-transitive closure is a preorder, not a partial order: mutuality relates co-nascent phenomena both ways (ch. 8 §29, including the base and the resultant mind at rebirth-linking, *paṭisandhikkhaṇe vatthuvipākā aññamaññaṃ*, `abh07t.nrf.xml:2913`). But its cycles all lie inside co-nascent clusters (a mind with its factors, the great elements of one cluster, the base with the resultant mind), none of which spans two streams, and every S edge raises the arising time, so the order it gives between streams is strict. "Preorder" describes the whole relation, not its cross-stream part. (3) Neither route makes S a causal set. As a lens only, this is relativity's shape rather than a causal set's: no global clock, yet a causal order between separate places, carried here by matter as there by signals. Nothing here bounds a speed, and the co-nascent cycles have no light-cone counterpart.

Co-nascence also presupposes a simultaneity, of a mind and the matter it originates, that a causal set lacks; a claim that one mind-moment occurs everywhere at once was checked and found no textual basis. For §3.5's third falsifier the result is mixed. S connects the kamma route, but the mind-born pair (*a*₂ before *b*₂) is an order the texts' own definitions imply and S alone cannot connect, so the falsifier is engaged in a narrowed form: S carries part of the order between streams, not all of it.

3.5 · What would falsify this row

- **A textual case in which a member of S holds with the conditioned arising at or before the condition** (for example, a canonical or commentarial passage in which proximity or pre-nascence runs from a later to an earlier phenomenon). S would then not be acyclic, and the surviving form fails.
- **A sub-commentarial or commentarial passage placing kamma's cross-moment conditioning on a present or future relatum.** This would remove kamma from S.
- **A temporal order the tradition asserts that S cannot connect.** If the tradition orders two phenomena in time, and no chain in S links them, then S does not carry the tradition's own temporal order, and "a strict order can be read over it" reduces to a formality. (§3.4 applies this test between streams and finds it engaged in a narrowed form: S carries the kamma-born route, and the mind-born pair is an implied order that S alone cannot connect. What remains open is a within-stream order that S cannot connect.)

---

4 · Rows 2 and 3 — Grain and Boundary: Where Space Is

4.1 · Physics primer: spin networks, and what "area has a discrete spectrum" means

Geometry is a field. In general relativity, the gravitational field *is* the geometry of spacetime: it decides lengths, areas and volumes. There is no stage apart from it. If the gravitational field is quantised, then lengths, areas and volumes become quantum quantities, and quantum quantities typically come in steps.

What "spectrum" means. In quantum theory, each measurable quantity is represented by an operator: a rule which, given the state of a system, yields the possible results of measuring it. The list of possible results is the quantity's **spectrum**. A spectrum is **discrete** if the possible results are separated by gaps. Think of a function that can only take values on the rungs of a ladder. An electron's energy in an atom has a discrete spectrum, and that is why atoms emit light at sharp colours. Loop quantum gravity finds that **area and volume** have discrete spectra (Rovelli and Smolin 1995; Ashtekar and Lewandowski 1997).

A spin network. A quantum state of space in loop quantum gravity is described by a **graph**: points (**nodes**) joined by lines (**links**). Each link carries a label j called a **spin**, taking values $j = ½, 1, 3/2, 2, \ldots$. Each node carries a second label, the **intertwiner**, which records how the links meeting there fit together.

```
              j₁
              │
       j₄ ────●──── j₂          ● a NODE: a grain ("quantum") of space
              │n                  each line: a LINK, labelled by a spin j
              j₃                  the node's volume depends on j₁…j₄ and the intertwiner
```

Area lives on links. Take any surface. Its area is a sum over the links that pierce it:

```
A  =  8πγ ℓ_P²  Σ  √( j (j + 1) )          sum over the links crossing the surface

ℓ_P  = √(ħG / c³)  ≈  1.6 × 10⁻³⁵ m       the Planck length
γ    = the Barbero–Immirzi parameter, a dimensionless constant of the theory
       (γ ≈ 0.2375 from black-hole entropy counting, Domagala–Lewandowski
        and Meissner 2004; ≈ 0.274 under the later Ghosh–Mitra counting;
        the value depends on the state-counting convention)
```

This is the formula for links that cross the surface transversally. When a node lies on the surface itself, the general formula adds further terms (Ashtekar and Lewandowski 1997). The smallest non-zero area is one link with $j = ½$:

```
A_min  =  8πγ ℓ_P² √(3/4)  =  4√3 π γ ℓ_P²   ≈  21.8 γ ℓ_P²
```

There is nothing between zero and this value. That gap is what "area has a discrete spectrum" means.

Volume lives at nodes. Each node has a volume, but volume is not an independent label. It is **computed** from the spins of the links meeting there and from the intertwiner, and it takes discrete values. A node where only three links meet has **zero** volume: the volume operator vanishes on gauge-invariant two- and three-valent nodes (Loll 1995 for the three-valent case, restated in Loll 1996, which computed the four-valent spectrum). In the geometric picture below, the same fact reads simply: three faces whose area-vectors sum to zero must lie in one plane, and a flat shape has no volume. Volume needs at least four links.

Nodes are quanta of space, not things in space. Rovelli (Zakopane lectures, §I.A): the quanta of gravity "do not live in space, but give rise to space themselves." There is no background in which the network sits. "Where" means "which node, adjacent to which".

Grains and faces. Bianchi, Doná and Speziale (2011) showed that a node can be read as a small polyhedron, a *quantum polyhedron*, whose faces are its links. The reading is semiclassical: coherent spin-network states represent semiclassical polyhedra, which is most reliable at large spins and least trustworthy at small spins, including the $j = ½$ where the smallest area sits. Adjacency is also definite only for a spin-network basis state on a fixed graph; a general quantum state of space is a superposition of graphs. Rovelli states the consequence plainly (Zakopane lectures, §II.F): "Two grains n and n' are adjacent if there is a link l connecting the two, and in this case the area of the elementary surface separating the two grains is determined by the spin of the link joining n and n'."

```
┌──────────────┐ ┌──────────────┐
│   grain  n   ├─┤   grain  n′  │       the face they share IS the link l;
│   (volume)   │ │   (volume)   │       its area = 8πγℓ_P² √(j_l (j_l + 1))
└──────────────┘ └──────────────┘

the same thing as a graph:     n ●──────────── j_l ────────────● n′
```

In one sentence: in loop quantum gravity, the boundary between two neighbouring grains of space is the adjacency itself, and it carries a quantised area.

4.2 · The Abhidhamma side: the grain

The kalāpa is matter. The smallest material cluster (*rūpa-kalāpa*) is not in the canonical Abhidhamma books. The *Dhammasaṅgaṇī* has no kalāpa vocabulary. The clusters appear in the commentary (the *Visuddhimagga* names the "pure octad", *suddhaṭṭhaka*, and the decads), and the manual systematises them into twenty-one kinds (*Abhidhammatthasaṅgaha* ch. 6: "*kalāpā ekavīsati*"). The simplest, the pure octad, is eight inseparable material phenomena: the four great elements (extension, cohesion, heat, motion) together with colour, odour, taste and nutritive essence.

The commentarial size ladder. The *Vibhaṅga* commentary (*Sammohavinodanī*), in a passage teaching a monk to see small faults as large, gives a ladder of sizes:

> *chattiṃsa paramāṇavo ekassa aṇuno pamāṇaṃ. Chattiṃsa aṇū ekāya tajjāriyā pamāṇaṃ. Chattiṃsa tajjāriyo eko rathareṇu.*
>
> Thirty-six *paramāṇu* make the measure of one *aṇu*; thirty-six *aṇu*, one *tajjārī*; thirty-six *tajjārī*, one *rathareṇu* [a mote of chariot dust].

So one *paramāṇu* is $1/36^3 =$ **1/46,656** of a chariot-dust mote. The commentary describes the *paramāṇu* as *ākāsakoṭṭhāsika*, of the order of a portion of space, and says it is visible only to the divine eye. **The ladder is for the *paramāṇu*, not the kalāpa.** The commentary does not use the word kalāpa there.

The identification of the two is a sub-commentary's, and it is contested. An earlier version of this section called it "later exegesis, common in modern summaries". That was wrong. Modern summaries repeat the identification, but they did not originate it. The *Visuddhimagga Mahāṭīkā* makes the identification, as the reason the powder mode of the four-elements meditation is taught:

> *Te pana kalāpā paramāṇuparimāṇā hontīti tappariyāpannānaṃ pathavīdhātuādīnaṃ ekasmiṃ sarīre parimeyyaparicchedaṃ dassetuṃ cuṇṇato manasikātabbatā vuttā.*
>
> Because those kalāpas are of *paramāṇu* size, attention by way of powder is taught, in order to show the measurable delimitation, in one body, of the earth element and the rest contained in them. (`e0103n.att.xml:6229`)

The same work reports a different ladder, of successive eighths, as the view of "some" (*keci*), and sets it aside for the commentary's (`e0103n.att.xml:6257`). Ledi Sayadaw declines the identification by name:

> *Yaṃ pana mahāṭīkāyaṃ te pana kalāpā paramāṇuparimāṇā hontīti vuttaṃ. Taṃ dhātūnaṃ cuṇṇamanasikārānurūpamattavasena vuttanti gahetabbaṃ.*
>
> What is said in the *Mahāṭīkā*, "those kalāpas are of *paramāṇu* size", is to be taken as said merely in conformity with the powder-attention to the elements. (*Paramatthadīpanī*, `e0301n.nrf.xml:4705`)

His own reason bounds the kalāpa from above: the embryo at rebirth-linking, three kalāpas' measure, "would be smaller even than a *paramāṇu*", because a *paramāṇu*, occurring by continuity as a mass of kalāpas accumulated moment by moment, would be "at the lower limit, of the measure of forty-nine kalāpas" (*heṭṭhima koṭiyā ekūnapaññāsakalāpaparimāṇoeva siyā*, `:4705`). So one kalāpa is at most a forty-ninth of a *paramāṇu*, with no lower size given. His *Anudīpanī*, which its colophon makes his own work (`e0401n.nrf.xml:1725`–`:1729`), compresses this to "a *paramāṇu* is of the measure of forty-nine kalāpas", and adds a harmonizing reading on which the *Mahāṭīkā*'s line "is also fitting as said taking the accumulated measure of the embryo" (*kalalassa vā upacitappamāṇaṃ gahetvā vuttantipi yujjati*, `e0401n.nrf.xml:1261`).

The tradition's own bridge from time to count to size. A sub-commentary on the *Abhidhammāvatāra* says of the eye-sensitivity that its own measure is "very subtle, with the *paramāṇu* as its limit" (*paramāṇupariyantena pamāṇena atisukhumameva*), and that it occurs in forty-nine kalāpas together, "because material phenomena have a life of seventeen mind-moments, and because kamma-born matter arises at each of the three sub-moments of every mind" (`abh08t.nrf.xml:7373`). Seventeen times three is fifty-one. The step from fifty-one to forty-nine (fifty-one groups present at any sub-moment, forty-nine of them at their presence phase) is this paper's arithmetic, checked in `d6_present_window`, `d6_standing_window` and `d6_counts_at_100`. It agrees with the *Vibhāvinī*'s forty-nine for decay (`abh07t.nrf.xml:4837`), and the *Vibhaṅga Anuṭīkā* reaches forty-nine by adding the dissolution-born to a series of forty-eight (`abh02t.tik.xml:4921`). **The count follows from time, and the size does not.** §4.8 takes up what happens if the size is read as a length.

4.3 · The Abhidhamma side: the boundary, in four layers

Canon — a cavity. The *Dhammasaṅgaṇī* (§637) defines the space element:

> *Katamaṃ taṃ rūpaṃ ākāsadhātu? Yo ākāso ākāsagataṃ aghaṃ aghagataṃ vivaro vivaragataṃ asamphuṭṭhaṃ catūhi mahābhūtehi — idaṃ taṃ rūpaṃ ākāsadhātu.*
>
> What is the material phenomenon that is the space element? That which is space, belonging to space, void, belonging to the void, an opening, belonging to an opening, **untouched by the four great elements** — this is the space element.

The *Vibhaṅga*, also canon, gives examples of the internal space element: the ear-holes, the nostrils, the mouth, the passage by which food is swallowed, where it rests and where it leaves. **At the canonical layer, space is a cavity with extent**: a gap in matter, the kind of thing that has room in it.

Canon — not unconditioned. The *Kathāvatthu* (VI.6, §460–462, verified in the Chaṭṭha Saṅgāyana text) refutes the thesis "*ākāso asaṅkhato*", that space is unconditioned. One of its arguments is homely: "*Udapāne khaññamāne anākāso ākāso hotīti? Āmantā.*" When a well is dug, does what was not space become space? Yes. And when an empty jar is filled, space disappears. Something that can be made and unmade is not unconditioned. The commentary attributes the rejected thesis to the Uttarāpathakas and the Mahisāsakas (so spelled there). The Mahiṃsāsakas are one of the four Vibhajjavādin branches in Cousins's account (Cousins 2001), which the sibling paper uses. Once again, the branch that kept the Pāli canon argues against a sibling branch.

Commentary — three spaces, and a characterisation. The *Kathāvatthu* commentary distinguishes three spaces:

> *tividho ākāso — paricchedākāso, kasiṇugghāṭimākāso, ajaṭākāso … Tesu paricchedākāso saṅkhato, itare dve paññattimattā.*
>
> Space is threefold: delimiting space, the space left by removing a meditation disk, and unentangled [open] space … Of these, **delimiting space is conditioned; the other two are mere concepts.**

The commentary's classing of open space as a concept is disputed within the tradition: the *Atthasālinī* reads the *Dhammasaṅgaṇī*'s space element as unentangled space (§7).

The *Visuddhimagga* (ch. XIV, CST ¶442) characterises the space element in its fourfold formula:

> *Rūpaparicchedalakkhaṇā ākāsadhātu, rūpapariyantappakāsanarasā, rūpamariyādapaccupaṭṭhānā, asamphuṭṭhabhāvacchiddavivarabhāvapaccupaṭṭhānā vā, paricchinnarūpapadaṭṭhānā. Yāya paricchinnesu rūpesu idamito uddhamadho tiriyanti ca hoti.*
>
> The space element has the characteristic of delimiting matter; its function is to display the boundaries of matter; it is manifested as the confines of matter, or as the state of being untouched, as holes and openings; **its proximate cause is the matter delimited.** It is that by which one can say of delimited material things, "this is above, below, around that."

The last sentence matters for §4.5. Direction and position are defined only **relative to delimited matter**.

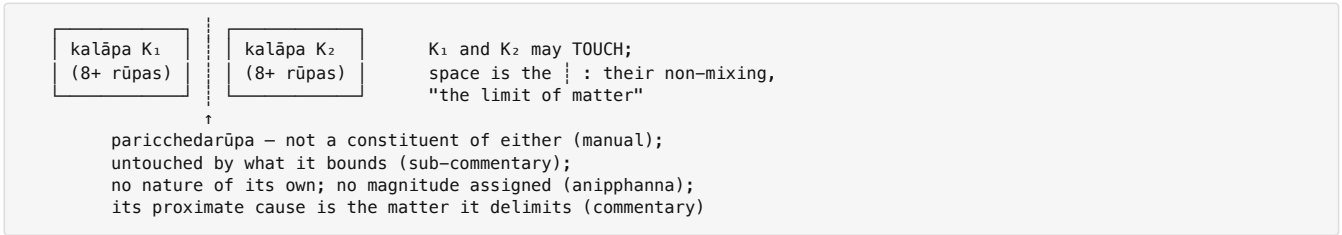
Manual — not a constituent. The *Abhidhammatthasaṅgaha* (ch. 6, the verse closing the section on kalāpas):

> *Kalāpānaṃ pariccheda-lakkhaṇattā vicakkhaṇā; Na kalāpaṅgamiccāhu, ākāsaṃ lakkhaṇāni ca.*
>
> Because space has the characteristic of delimiting the kalāpas, and because of the nature of the characteristics, the wise do not call space, nor the characteristics, a constituent of a kalāpa.

Space is **between** kalāpas, never **in** one. The manual also classes this delimiting matter (*paricchedarūpa*) as **anipphanna**: not concretely produced, with no intrinsic nature of its own. The term is older than the manual; the *Visuddhimagga* and the *Sammohavinodanī* already use it. It is counted as one of the twenty-eight material phenomena, but the standard expositions do not count it among the ultimate realities (Bodhi 1993, ch. VI), and one reads it as "nothing but the mere limitation of matter" (Karunadasa 1967, ch. 5).

Sub-commentary — the non-mixing of touching neighbours. The *Vibhāvinī*, on the same chapter:

> *Cakkhudasakādiekekakalāpagatarūpānaṃ kalāpantarehi asaṃkiṇṇabhāvāpādanavasena paricchedakaṃ, tehi vā paricchijjamānaṃ, tesaṃ paricchedamattaṃ vā rūpaṃ paricchedarūpaṃ … Vijjamānepi ca kalāpantarabhūtehi kalāpantarabhūtānaṃ samphuṭṭhabhāve taṃtaṃrūpavivittatā rūpapariyanto ākāso. Yesañca so paricchedo, tehi sayaṃ asamphuṭṭhoyeva. Aññathā paricchinnatā na siyā tesaṃ rūpānaṃ byāpībhāvāpattito. Abyāpitā hi asamphuṭṭhatā.*
>
> Delimiting matter is what delimits the matter in each kalāpa (the eye-decad and the rest) by making it **unmixed with neighbouring kalāpas**; or what is delimited by them; or their mere delimitation … **Even where neighbouring kalāpas touch one another, space is the separateness of each one's matter, the limit of matter.** And it is itself untouched by the things it delimits. Otherwise those material things would not be delimited, since they would come to pervade one another. For non-pervasion is non-touching.



4.4 · An older neighbour: do minimal units touch? *(cross-tradition: Sarvāstivāda, in Sanskrit)*

The *Vibhāvinī*'s wording answers a question the Indian schools had argued for centuries. The *Abhidharmakośabhāṣya* (ad I.43, Pradhan ed., pp. 32–33) asks it directly:

> *kiṃ punaḥ paramāṇavaḥ spṛśanty anyonyam āhosvin na / na spṛśantīti kāśmīrakāḥ / … yadi tāvat sarvātmanā spṛśeyur miśrībhaveyur dravyāṇi / athaikadeśena sāvayavāḥ prasajyeran / niravayavāś ca paramāṇavaḥ*
>
> Do atoms touch one another, or not? The Kashmiri masters say they do not touch. … If they touched with their whole selves, the substances would **mix**. If they touched with a part, they would have parts. But atoms are partless.

The same passage adds that contact means only arising *without interval*, and that "their being without interval is just that there is nothing between them" (*tad evaiṣāṃ nirantaratvaṃ yan madhye nāsti kiñcit*). The form *nirantaratvaṃ* is this paper's emendation: the GRETIL e-text reads *niruttaratvaṃ* [032|20]. The emendation follows the passage's own *nirantara* in the line before, and Pradhan's printed page was not checked. Bhadanta Vasumitra holds that they do not touch, and that we call it touching when there is no interval.

Two cautions before the comparison. First, the *Kośa*'s contact debate concerns **partless atoms** (*paramāṇu*). Its functional counterpart of the kalāpa is a different item, the smallest aggregate of eight substances (*aṣṭadravyaka*, ad II.22), the counterpart of the pure octad. So the comparison below is of the **role** of minimal units, and it does not identify the *Kośa*'s atom with the *Sammohavinodanī*'s *paramāṇu* (which §4.2 already keeps apart from the kalāpa). Second, the *Kośa* also has a conditioned space element. Besides unconditioned *ākāśa* (I.5), it names the cavity: *dvāravātāyanamukhanāsikādiṣu chidram ākāśadhātvākhyam* (I.28, [018|13]), "the hole in a door, a window, the mouth, the nose and so on is called the space element", which it says is light and darkness (*ālokatamasī kila*). That is a close parallel to the *Vibhaṅga*'s ear-holes and nostrils (§4.3).

The two traditions share the premise: **if minimal units truly touched, they would mix.** The *Vibhāvinī* keeps the premise by making touching conventional only: "non-pervasion is non-touching" (*abyāpitā hi asamphuṭṭhatā*). They draw different conclusions:

	Do the units touch?	**What is between them?**
Kashmiri Sarvāstivāda (*Kośa* ad I.43)	no	**nothing** ("nothing between them")
Theravāda sub-commentary (*Vibhāvinī*)	conventionally yes; strictly no (non-pervasion is non-touching)	**something**: delimiting matter, the non-mixing — counted among the material phenomena, though *anipphanna* and not an ultimate reality
Loop quantum gravity	grains share a face	**something**: the link, which carries area

On this one question, the Theravāda answer and loop quantum gravity have the same *shape*: the between is not nothing. It is a named, counted item of the list (one of the twenty-eight material phenomena in one, a link in the other), and it is not itself one of the minimal units: not a kalāpa, not a node. The Kashmiri answer makes the between empty. Two limits on that shape. The Theravāda item is counted but is not an ultimate reality, and Karunadasa (1967) reads it as the mere limitation of matter, which moves it toward the Kashmiri *nothing*: what separates the two answers on that reading is that one names and counts the between and the other does not. And the two "neither" relations are opposite in incidence (§4.6, break 6). The Sarvāstivāda counts *ākāśa* among the unconditioned (I.5), which is the thesis the *Kathāvatthu* refutes, and separately names the conditioned cavity (I.28).

4.5 · The correspondence, and its surviving form

> **Both frameworks put the unit of spatial boundary at the adjacency between minimal units.** In loop quantum gravity, a spin-network link is the face that two neighbouring grains of space share. In the *Vibhāvinī*, space is the non-mixing of touching neighbouring kalāpas, "the limit of matter", untouched by what it bounds and a constituent of neither. In neither framework is the boundary itself one of the minimal units. In both, "where" is defined by adjacency and not by a background: Rovelli's quanta "do not live in space", and the *Visuddhimagga*'s space is "that by which one can say of delimited material things, 'this is above, below, around that'".

That is the whole of the claim. The next paragraphs give the tradition's reason for it; everything else in this section limits it.

Why the between is matter's, and never mind's. The tradition gives the reason, from the commentary on. The *Paṭṭhāna* commentary calls successive mind-moments contiguous "thoroughly without interval, because of the absence of shape" (*saṇṭhānābhāvato suṭṭhu anantarāti samanantarā*, `abh03a.att.xml:7697`). Its *Mūlaṭīkā* spells this out: mental phenomena, lacking shape as material phenomena have it, and with condition and conditioned never standing together, admit no division "this is below, above, across from that" (*rūpadhammānaṃ viya saṇṭhānābhāvato paccayapaccayuppannānaṃ sahāvaṭṭhānābhāvato ca "idamito heṭṭhā uddhaṃ tiriya"nti vibhāgābhāvā*, `abh03t.tik.xml:4305`), which is the *Visuddhimagga*'s formula for what space does for matter (§4.3). Its *Anuṭīkā* ties the absence of division to shapelessness (`abh05t.nrf.xml:4637`), and the *Vibhāvinī* defines contiguity by it (*saṇṭhānābhāvena suṭṭhu anantarapaccayo samanantarapaccayo*, `abh07t.nrf.xml:6653`). Ledi Sayadaw draws the contrast in so many words: between two material clusters, even bound together in a stone pillar, "there is an interval, an opening; not so between the earlier and later two groups of mind and mental factors", which, having no shape, have no delimiter and no interval at all (*dvinnaṃ rūpakalāpānaṃ majjhe antaraṃ nāma vivaraṃ nāma atthiyeva. Na tathā purimapacchimānaṃ dvinnaṃ cittacetasika kalāpānaṃ majjhe*, *Paṭṭhānuddesa-dīpanī*, `e0501n.nrf.xml:229`).

The two reasons reach differently. Within one stream both hold: no two moments are present at once ("even for Buddhas, two minds do not occur together", *buddhānampi hi dve cittāni ekato nappavattanti*, *Suttanipāta* commentary, `s0505a.att.xml:685`), and what is present together in one moment, consciousness with its factors, is conjoined, "one in arising and ceasing, one in object and base" (*ekuppādanirodhā ca, ekālambaṇavatthukā*, manual ch. 2 §1, `abh07t.nrf.xml:329`), the opposite of the non-mixing that defines the delimiter. Across streams,

where moments of two minds can be present at once, the second reason does not apply and shapelessness must do the work; these passages speak of succession within a stream and do not discuss that case.

Only the within-stream precondition has been checked by machine, in Lean 4, in the public repository `SiliconWat/formal-abhidhamma` (CC0; software concept DOI 10.5281/zenodo.23067241), file `kala/Time.lean` with the definitions in `kala/Axioms.lean` (commit `deb9dec`):

```
a stream s:   moments, a successor map, and a clock  tick : Moment → ℤ
              one_at_a_time :  tick a = tick b  →  a = b              (A1: tick is injective)
a kalāpa k:   birth : ℤ ;   k present at t  :=  birth ≤ t < birth + 17
between k₁ k₂ t  :=  k₁ present at t  ∧  k₂ present at t              (co-presence only)

theorem mind_has_no_between (s) (t) :   ¬ ∃ a b,  a ≠ b ∧ tick a = t ∧ tick b = t
theorem matter_can_have_a_between :     ∃ k₁ k₂,  k₁ ≠ k₂ ∧ between k₁ k₂ 5    (a witness)
```

What the check reaches is small. `mind_has_no_between` is the one-at-a-time axiom restated as the injectivity of `tick`; it never mentions `between`, which is defined only on kalāpas and only as co-presence. Kalāpas carry a birth on an integer line in mind-moment units and are attached to no stream. Nothing in the model touches, adjoins or has a shape, so neither the row's adjacency nor the shapelessness reason is formalised. The run script, re-run for this revision, checks these two and five more theorems in the file, and each of six deliberate breaks of the axioms (among them dropping the one-at-a-time axiom) makes the build fail. The file is exploratory, outside the repository's pre-registration, and its model is the authors'. The asymmetry is also a known shape: a stream is a chain (§3.1), and kalāpas present together play the part of an **antichain**, a set of elements none of which precedes another; that the between belongs to the antichain adds nothing specific to loop quantum gravity (§9).

The grain row survives only through the boundary row. Kalāpa against node, taken alone, is a comparison of a smallest-unit-of-matter with a smallest-unit-of-space. That is the general "smallest unit" comparison the census found common, and it fails on the matter/geometry distinction below. It earns a line in the table only because the boundary row presupposes minimal units with neighbours.

4.6 · Where it breaks

1. **The dependence runs in opposite directions.** In loop quantum gravity, the adjacency data come first: the spins on the links, with the intertwiners, determine the volumes of the nodes. In the Abhidhamma, the matter comes first: the space element's **proximate cause is the matter delimited** (*paricchinnarūpapadaṭṭhānā*, commentary). One builds the grain from its boundaries. The other derives the boundary from its grains.
2. **Magnitude.** Loop-gravity area is observable and has a discrete spectrum, with a smallest non-zero value. Abhidhamma delimiting matter is *anipphanna*: it has no nature of its own, and no magnitude is assigned to it. ("No magnitude" is this paper's reading of *anipphanna*; no text says "it has no size" in those words.) There is no "how much" boundary between two kalāpas. The canonical cavity (break 4) does have extent, but no text measures it. The Abhidhamma boundary is object-discrete (it is one item between two items), but not spectrally discrete (§2).
3. **Matter against geometry.** Kalāpas are matter *in* space. Their octad includes colour, odour and taste. Spin-network nodes are quanta *of* space and carry no qualities except geometry. The commentarial size ladder (§4.2) measures the *paramāṇu* against a chariot-dust mote, which presupposes a background of lengths. That is exactly the background loop quantum gravity denies.
4. **Layer.** The correspondence holds at the **sub-commentary** layer. At the **canonical** layer, space is a cavity with extent: ear-holes, a well being dug, an opening. A cavity is volume-like, not face-like. So the canon's space corresponds, if to anything, to a region, not to a link. The delimitation reading enters with the commentary's characterization of the space element as delimiting matter (§4.3), and the between-kalāpas reading is completed only in the manual and sub-commentary, more than a millennium after the canon was compiled.
5. **Open space.** The *Kathāvatthu* commentary calls open space (*ajaṭākāsa*) a mere concept. The *Atthasālinī* reads otherwise: on its reading, the *Dhammasaṅgaṇī*'s space element is unentangled space, a material phenomenon (§7). Loop quantum gravity has no space without the network: a region with no nodes has no geometry at all. On the *Kathāvatthu* commentary's reading the two may look alike here, but the likeness is loose. One says that open space is a *designation*; the other says there is *nothing there to designate*. On the *Atthasālinī*'s reading there is no likeness at all, since open space is then a material phenomenon.
6. **Opposite incidence.** A spin-network link is a face **incident to both** grains: each grain's polyhedron has it as one of its own faces, and the area belongs to the geometry of both. The *Vibhāvinī*'s delimiter is **untouched by both** kalāpas (*tehi sayaṃ asamphuṭṭhoyeva*) and a constituent of neither. "Belongs to neither neighbour" covers both only because it is vague. Said exactly, the link is shared and the delimiter is excluded, which are opposite relations. What the two have in common is only that the boundary is not itself a minimal unit.
7. **Counted, but not ultimate.** The delimiter is one of the twenty-eight material phenomena, but it is not an ultimate reality (Bodhi 1993), and on Karunadasa's reading (1967) it is "nothing but the mere limitation of matter". The link in loop quantum gravity is as real as

anything in the theory. So the Theravāda "something" between kalāpas is a *named* something, not a real unit, and the row's contrast with the Kashmiri *nothing* rests on the naming and counting.

4.7 · What would falsify this row

- **A text, at any layer, that makes delimiting matter a constituent of a kalāpa**, or places space inside a cluster rather than between clusters. The manual's verse would then be contradicted within the tradition, and the "between" would lose its textual footing.
- **A formulation of loop quantum gravity in which the area between adjacent grains is not carried by their adjacency.** For example, the polyhedral reading may not hold at small spins, since it is a semiclassical, coherent-state interpretation; or area might attach to nodes rather than links. The physics side of the correspondence would then go.
- **A text on which two moments of one mental stream are present at once, on which mental phenomena have shape, or on which space delimits mental phenomena.** The tradition's reason in §4.5 (matter has a between, mind has none) would then fail, and the Lean check would describe the authors' model and not the texts.
- **A reading of the *Vibhāvinī* on which "space" is a gap with nothing in it** (the Kashmiri answer), rather than a counted *rūpa*. The §4.4 table would then collapse, and the row would reduce to the generic "space is between" of §9. This falsifier is **partly engaged** already. The *Vibhāvinī*'s own third gloss is "their mere delimitation" (*tesaṃ paricchedamattaṃ*), and Karunadasa reads the *anipphanna* class as "nothing but the mere limitation of matter". What keeps the row from collapsing is only that the tradition still lists and counts the item. If that counting were shown to be nominal and nothing more, the falsifier would fire.

4.8 · The grain's one textual size, carried outward

The rows above compare shapes. This one puts a number on the grain, because the *Mahāṭīkā*'s line (§4.2) invites it, and the result is a collision. The comparison needs four steps the texts do not take, all of them this paper's: an *aṅgula* (finger-breadth) in meters, taken here as 1.7–2.5 cm; the ladder read as metrology, though it was taught as a device for seeing small faults as large (`abh02a.att.xml:6333`); a uniform size carried from bodily matter to external matter, though every textual size concerns the body (the *Mahāṭīkā*'s "in one body", Ledi's embryo, the sub-commentary's eye); and "forty-nine" read one way or another.

The ladder gives $7^3 \times 36^4 = 576{,}108{,}288$ *paramāṇu* to the *aṅgula*, so one *paramāṇu* is about 3–4 $\times$ 10^{-11} m (`d7_ladder`, `d7_sizes`). Rutherford's 1911 scattering law, which treats the central charge as a point, has an α particle reaching about 3.4 $\times$ 10^{-14} m, or 34 femtometers (fm), of the center; the proton's charge radius is 0.84 fm (Pohl et al. 2010); Hofstadter's electron scattering (1956) maps charge distributions at the femtometer scale.

READING OF ONE KALĀPA	SIZE, fm	AGAINST 34 fm	AGAINST 0.84 fm
Mahāṭīkā: kalāpa = paramāṇu	29,500 – 43,400	≈ 870 – 1,280 ×	≈ 35,000 – 52,000 ×
"49" as an equality, read by volume	8,100 – 11,900	≈ 240 – 350 ×	≈ 9,600 – 14,000 ×
"49" as an equality, read linearly	600 – 890	≈ 18 – 26 ×	≈ 720 – 1,050 ×
Ledi as he wrote it: at most paramāṇu/49, no lower size	no size	no collision	no collision

The size collision has a weakness of its own: if magnitudes are designations (§2), a clash of magnitudes might be set aside. **The count does not have that weakness.** A gold nucleus, about 7 fm in radius and so wholly inside one kalāpa on every sized reading above, holds 197 individuated nucleons. A kalāpa of external matter is a pure octad, with eight members, and the plurality of dhammas is what the teachers' method leaves real. If what is individuated inside a kalāpa must be one of its members (this paper's premise), 197 cannot fit in eight (`d7_count_collision`). No anchor in the stated range clears this.

The ladder also misses its own visible rung. The commentary says the *aṇu* (thirty-six *paramāṇu*) appears whirling in a sunbeam through a crack (`abh02a.att.xml:6333`). On the same anchors an *aṇu* is about 1–1.6 nm, some six to nine hundred times smaller than the roughly one-micrometer dust that makes a visible sunbeam (the threshold is a rough figure of this paper's). "Swirls" (*vaṭṭi vaṭṭi*) describes groups, not one *aṇu*, and nanometer particles still do not form them. This extends the comparison the formal-abhidhamma run *the outside world* made at the ladder's louse and nit rungs.

Verdict: a collision with a reading, again. The *Mahāṭīkā*'s line, taken as the metric size of one kalāpa and carried to external matter, collides with measured nuclear structure. Taken as Ledi takes it, it does not, and then **no grain of extent for matter follows from the texts**. Space has a smallest portion only as a designation: the commentary's *paramāṇu* is *ākāsakoṭṭhāsika*, and the *Anudīpanī* glosses that as placed by people in a space-portion (*manussehi ākāsakoṭṭhāse ṭhapito*, `e0401n.nrf.xml:1261`). Ledi's exit is the tradition's own and was not built for this comparison: he wrote before these measurements were made. It leaves the row weaker, which fits §4.5: the grain survives as what the boundary presupposes, with no length of its own.

What would falsify this subsection. A text giving a kalāpa of external matter a size; a text making the *Mahāṭīkā*'s line a measurement and not a teaching device; or a reading on which what is individuated in a nucleus is no plurality of dhammas at all, which would remove the count collision. The last is easy to supply, and that is a limit of the collision, stated in §12.

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5 · Row 4 — Ratio: Time as a Count

5.1 · Physics primer: a duration is a number of steps

Recall from §3.1 that in a causal set the proper time between two related events is proportional to the length of the longest chain between them. There is no clock apart from counting. Two durations can be compared by comparing chain lengths, and the ratio is meaningful even though no unit of time is ever introduced. (Converting to seconds needs the sprinkling density, one element per Planck volume. The ratio does not need it.)

5.2 · The Abhidhamma side

Three sub-moments per mind-moment. The manual (ch. 4 §8, `abh07t.nrf.xml:1229`): "*Uppādaṭhitibhaṅgavasena khaṇattayaṃ ekacittakkhaṇaṃ nāma*" — a mind-moment consists of three sub-moments: arising, presence, dissolution.

Seventeen mind-moments per material phenomenon. The next line: "*Tāni pana sattarasa cittakkhaṇāni rūpadhammānamāyū*" — seventeen such mind-moments are the lifespan of material phenomena. The manual then shows where seventeen comes from. It is the length of the longest process by which a sense object is known: one past life-continuum moment, two vibrations of the life-continuum, and fourteen moments of the process itself: "*cuddasa vīthicittuppādā, dve bhavaṅgacalanāni, pubbevātītakamekacittakkhaṇanti katvā sattarasa cittakkhaṇāni paripūrenti*" (§11, `:1241`). **Seventeen is a consistency condition.** A physical object must last as long as the longest process that cognises it.

The commentary gives the same figure in its own words: while matter persists, sixteen minds arise and cease, and it ceases with the seventeenth (*rūpe dharanteyeva soḷasa cittāni uppajjitvā nirujjhanti. Taṃ pana sattarasamena cittena saddhiṃ nirujjhati*, *Sammohavinodanī*, `abh02a.att.xml:413`). Its simile has a short-legged and a long-legged man walking together, sixteen steps to one, and its lesson is stated first: "matter cannot go on leaving mind behind, nor mind leaving matter; the going of both is of one measure" (*rūpaṃ pana arūpaṃ arūpaṃ vā rūpaṃ ohāya pavattituṃ na sakkonti. Dvinnampi ekappamāṇāva pavatti*, `abh02a.att.xml:417`).

Fifty-one sub-moments, with exceptions. The sub-commentary: "*Visuṃ visuṃ pana ekapaññāsa cittakkhaṇāni honti. Rūpadhammānanti viññattilakkhaṇarūpavajjānaṃ rūpadhammānaṃ*" — counted separately, they are fifty-one; "material phenomena" means those other than the intimations and the characteristic phenomena (`abh07t.nrf.xml:4837`). It then states the exceptions. The two intimations (bodily and vocal) last one mind-moment. Of the characteristics, arising and impermanence match the mind-moment's arising and dissolution sub-moments, and decay lasts forty-nine sub-moments. **There is no single material clock.**

Seventeen, sixteen and a third, or sixteen: the count is disputed. The *Sammohavinodanī* holds two rules. It first dates kamma-born matter by one stated "because it came in the [older] commentary" (*aṭṭhakathāyaṃ āgatattā vuttaṃ*): matter arisen at one mind's arising sub-moment ceases at another's (`abh02a.att.xml:421`), sixteen mind-moments and a third. It then rejects that rule as conflicting with the canonical *Yamaka* (*imāya pāḷiyā virujjhati*, `:429`) and replaces it with seventeen: matter arisen with the rebirth-linking mind ceases with the seventeenth, and matter arisen at that mind's presence sub-moment ceases at the arising of the eighteenth (`:433`). The *Vibhaṅga Mūlaṭīkā* reads the passage so (`abh02t.tik.xml:305`), and finds both the seventeen and the older "sixteen and a third" (*yā ca aṭṭhakathāyaṃ tatiyabhāgādhikasoḷasacittakkhaṇāyukatā*, `:309`) at odds with the commentary's own account of death and rebirth-linking, in which eleven mind-moments of an object's life have passed and five remain (`abh02a.att.xml:2889`; `e0102n.mul.xml:4045`). Its *Anuṭīkā* concludes "sixteen only" (`abh02t.tik.xml:4933`), and four lines later speaks of "sixteen or seventeen" (*soḷasahi sattarasahi vā cittakkhaṇehi*, `:4937`). The *Vibhāvinī* reports the sixteen as the view of "some" (*keci*) and rejects it as without substance (*tayidamasāraṃ*), citing the replacement rule, and reads any sixteen as a count of the moments fit to condition mind (`abh07t.nrf.xml:4845`); Ledi Sayadaw finds that rejection "not good" (*taṃ na sundaraṃ*, `e0301n.nrf.xml:2745`). Seventeen is the manual's figure and the *Vibhāvinī*'s, not a settled one.

The presence sub-moment is disputed too. The *Vibhāvinī* reports that "some reject the presence moment of consciousness" (*keci pana cittassa ṭhitikkhaṇaṃ paṭisedhenti*), because the canonical *Yamaka* names only arising and dissolution (*bhaṅguppādāva kathitā, na ṭhitikkhaṇo*, `abh07t.nrf.xml:4833`). It names no one; the cross-reference to the *Vibhaṅga Mūlaṭīkā* beside the passage is the edition's. The *Mūlaṭīkā* does raise the question in its own voice, on the same ground: "whether it exists or not is to be examined" (*so ca atthi natthīti vicāretabbo*, `abh02t.tik.xml:313`). Sumaṅgala **rejects** the dissent, citing the commentary's three sub-moments. Without presence, this paper's arithmetic gives $17 \times 2 = \mathbf{34}$ in place of 51 (the dissent states no count), and decay's forty-nine sub-moments would be undefined.

An absolute scale, named but not timed. The *Vibhaṅga* commentary says of feeling, a mental factor present in every consciousness, that "in the moment of a finger-snap it arises and ceases a hundred thousand *koṭi* times" (*ekaccharakkhaṇe koṭisatasahassasaṅkhyā uppajjitvā nirujjhati*, `abh02a.att.xml:533`). The texts say what a finger-snap *is* without giving it a duration. The *Aṅguttara* commentary names it by a gesture, "the mere making of a sound by striking two fingers" (*dve aṅguliyo paharitvā saddakaraṇamattanti attho*, `s0401a.att.xml:1119`). A Pāli grammar, the *Payogasiddhi*, makes it the unit of vowel length: "the short time is the time called a finger-snap or an eye-blink; by it the short vowel has one measure" (*rassakālo nāma accharāsaṅghāto akkhinimmilanasaṅkhāto vā kālo, tena ekamatto rasso*, `e0807n.nrf.xml:205`). Each fixes a **unit of reckoning**, not a number of seconds. This paper takes a finger-snap as about a tenth of a second, and that assumption is its own.

The texts state the rate of mind itself as **round lower bounds in four works**, not as one number:

Work (layer)	Locus	Words	Reading
Saṃyutta commentary, on SN 12.61 (commentary)	`s0302a.att.xml:1463`	*ekasmiñhi accharākkhaṇe anekāni cittakoṭisatasahassāni uppajjanti*	many × 10^{12} mind-moments per snap
Milindapañha, quoted on the next line of the same commentary	`s0518m.nrf.xml:2838`; `s0302a.att.xml:1467`	*ekaccharākkhaṇe pavattacittassa ettakā vīthī lakkhaṃ ṭhapīyamānā … parikkhayaṃ pariyādānaṃ gaccheyyuṃ*	more minds in a snap than grains in over a hundred cartloads of rice
Sāratthadīpanī, on the Vinaya (sub-commentary)	`vin01t2.tik.xml:341`	*accharāsaṅghātamatte khaṇe anekakoṭisahassacittappavattisambhavato*	many × 10^{10} per snap
Dhammasaṅgaṇī Anuṭīkā (sub-commentary)	`abh04t.nrf.xml:821`	*accharāsaṅghātakkhaṇena anekakoṭisaṅkhāya cittuppattiyā*	many × 10^{7} per snap

The readings take *koṭi* as 10^7, *satasahassa* as 10^5 and *sahassa* as 10^3; *aneka*, "not one", makes each a floor. Read strictly as "at least two", it would put the first floor at 2×10^{12}, above the commentary's own flat 10^{12} for feeling, which arises and ceases with its mind; so either *aneka* is not a strict floor of two, or two of Buddhaghosa's commentaries differ by a factor of two. This paper reads the figures as round orders of magnitude. None of the four works states a ceiling. On the tenth-of-a-second assumption, the floors put a mind-moment at or below about 10^{-13} to 10^{-8} seconds, far below any timescale on which perception is measured.

The texts deny that this succession is seen by the reasoner and by "others", not by everyone. The *Dīgha* commentary says the speculative reasoner of DN 1 "does not see the breaking up of mind" (*cittassa bhedaṃ na passati*), because each mind-moment ceases only after conditioning the next, and so takes mind to fly on elsewhere like a bird (`s0101a.att.xml:2245`): a diagnosis of his error. The *Sāratthadīpanī* says the order of a Buddha's teaching "is not discerned by others", while the order of several monks' questions "is discerned by one so wise" (*pucchanavāro tādisassa paññavato paññāyati … Buddhānaṃ pana desanāvāro aññesaṃ na paññāyateva*, `vin01t2.tik.xml:341`), and the *Vibhaṅga Anuṭīkā* reserves true knowledge of the elements to the Tathāgata (*nānādhātūsu tathāgatasseva yathābhūtañāṇaṃ*, `abh02t.tik.xml:4937`). So the tradition does give the mind-moment an outside scale, the snap, and the offer cannot be taken as a measurement. (Three of the four floors were supplied by the refuter of the formal-abhidhamma run *rate-of-mind control*, 2026-09-30, `runs/2026-09-30-rate-of-mind-control/RUN.md`.)

5.3 · The correspondence, and its surviving form

The Abhidhamma measures the lifetime of matter **in mind-moments**. This paper counts the lifetime by its **elements**, as the manual does (*sattarasa cittakkhaṇāni*, seventeen moments), because each mind-moment is itself a stretch of time. A causal-set chain is conventionally counted by its links, since its elements are points; the difference of one does not matter at large counts, but it does at seventeen.

Two cases, which the texts treat differently. An earlier version of this paper had matter "timed by the mind's chain". The texts support that where a stream exists, and not where none does.

- **Where a stream exists, its own mind-moments date its matter.** The *Sammohavinodanī* has kamma-born matter arise at each of the three sub-moments of every mind (*ekekassa cittassa tīsu tīsu khaṇesu samatiṃsa samatiṃsa kammajarūpāni uppajjanti*, `abh02a.att.xml:421`), dates mind-born and kamma-born matter from the mind it arises with to the seventeenth (`:429`–`:433`, §5.2), and ends the kamma-born series with the death-mind (*tato paṭṭhāya kammajarūpapaveṇī na pavattati*, `:425`). Mind dates this matter without conditioning it: after rebirth-linking, kamma-born matter goes on "without mind, from kamma alone" (*vinā cittena kammatova kaṭattārūpānaṃ pavatti hotīti*, *Paṭṭhāna* commentary, `abh03a.att.xml:7857`).
- **Where no stream exists, no text counts matter in mind-moments.** The mindless beings (*asaññasatta*) are matter without mind. At the canonical layer the *Dhātukathā* makes theirs an existence of **one** aggregate (*ekavokārabhavo ekena khandhena … saṅgahito*, `abh03m1.mul.xml:421`; the *Vibhaṅga* lists it, `abh02m.mul.xml:3685`), and the *Kathāvatthu* (III.11) refutes an opponent who read DN 1's line that these deities fall away "when perception arises" (*saññuppādā ca pana te devā tamhā kāyā cavanti*, `s0101m.mul.xml:487`) as

proof of perception among them (`abh03m3.mul.xml:5797`). The *Paṭṭhāna* commentary calls it the place of matter, where "material phenomena arise without even a rebirth-linking mind as a mental condition" (*paṭisandhicittamattampi arūpapaccayaṃ vinā rūpadhammāva uppajjanti*, `abh03a.att.xml:7861`). The manual gives them no occurrence of mind at all (*sabbathāpi cittappavatti natthevāti*, `abh07t.nrf.xml:1485`; also `:1957`, `:2349`) and a lifespan of five hundred aeons (`:1693`), and the *Dīgha* sub-commentary answers "for how long?" in aeons: "at most five hundred great aeons", by the momentum of the kamma, like an arrow from a bow (*ukkaṃsato pañca mahākappasatānipi tiṭṭhanti asaññasattā. Jhānavegeti asaññasamāpattiparikkhate kammavege*, `s0101t.tik.xml:2053`). Matter outside every being is the same case: temperature produces matter "internally and externally" (*ajjhattañca bahiddhā ca*, manual ch. 6 §35, `abh07t.nrf.xml:2213`), and what lies outside one's continuum is external "whether bound to faculties or not" (*indriyabaddhā vā anindriyabaddhā vā*, *Atthasālinī*, `abh01a.att.xml:3485`), like the two trees of §13.

So the general statement of seventeen is not tied to a present mind. The *Vibhāvinī* glosses the manual's "seventeen mind-moments" as "moments like the moments of seventeen minds" (*sattarasannaṃ cittānaṃ khaṇāni viya khaṇāni*, `abh07t.nrf.xml:4837`; likewise "one mind-moment", `:4849`), and Ledi Sayadaw calls the division of one moment into mind's and matter's a conceptual construction (*kappanāvasena*, `e0301n.nrf.xml:2749`). That is the reading of the seventeen as a **unit of reckoning**, a duration counted in mind-moment lengths. It is not the only reading open: the *Vibhaṅga Anuṭīkā* derives matter's longer life from its being made known by mind, in sixteen or seventeen mind-moments (`abh02t.tik.xml:4937`), which leaves the reading that the count describes matter as cognised, and says nothing of matter no mind cognises.

> **Where a stream exists, the texts date its mind-born and kamma-born matter on its own mind-moments, sub-moment by sub-moment, without making mind that matter's condition. Where none exists, no text counts matter in mind-moments: the mindless beings' material series is timed by kamma's momentum in aeons, and a sub-commentary makes the mind-moment a unit. What survives of the causal-set comparison is generic: a duration stated as a count of steps (§9).**

The row's name, *ratio*, needs one correction. The ratio is 17 : 1, a material lifetime against one mind-moment, but both terms are measured in the **same** unit. Matter has no chain of its own in S, so the row gives a ratio of durations, not a ratio of two chain lengths.

A second reckoning, and how the texts relate it. Asked how long an aeon (*kappa*) is, the Buddha answers that it is not easy to count in "so many years, or so many hundreds of years …" (*so na sukaro saṅkhātuṃ ettakāni vassāni iti vā, ettakāni vassasatāni iti vā*) and gives the simile of a mountain a league on each side, wiped with a cloth once a century, which would wear away before the aeon ends (SN 15.5, `s0302m.mul.xml:3529` – `:3533`; past aeons are "not easy to count", SN 15.8, `:3565`). The aeon is not counted in years; the simile sets it against a process timed in centuries. The texts do relate it to the mind-moment, in the mindless beings' case, and the relation is divergence. Since "the usage 'time' rests only on the occurrence of phenomena", the *Paṭṭhāna Mūlaṭīkā* finds "no time-interval" between the death before that existence and the rebirth-linking after it, for "the occurrence of matter makes no interval in the occurrence of mind, because it is another series" (*kālantaratā natthi … na ca rūpadhammappavatti arūpadhammappavattiyā antaraṃ karoti aññasantānattā*, `abh03t.tik.xml:4309`); its *Anuṭīkā* names the objection this answers, "how then is it said, 'having passed five hundred aeons'?" (`abh05t.nrf.xml:4641`). The *Paṭṭhāna* commentary had already rejected a view of "the teachers" that contiguity means nearness in time, because the attainment before cessation is still the contiguity condition of the fruition after it, and granted that "there is no nearness in time there" (*kālānantaratā natthi*, `abh03a.att.xml:7697`); the *Vibhāvinī* follows it and counts the rebirth-linking that arises "even after a time-interval" from the mindless existence the same way (*kālantarepi*, `abh07t.nrf.xml:6653`). Ledi Sayadaw: "between two minds, in the non-percipient existence, the person stands without mind for five hundred aeons" (*dvinnaṃ cittānaṃ antare asaññasattabhave pañcakappasatāni puggalo acittako tiṭṭhati*, `e0501n.nrf.xml:233`). The two reckonings meet and do not agree: five hundred aeons in one, no interval in the other. As a lens only, the pair recalls relativity's proper time along a worldline and the cosmic time of the expanding (FLRW) cosmologies; nothing in the likeness is a prediction. For this row the divergence settles one point: the aeon is not a second chain for the seventeen to be compared with.

5.4 · Where it breaks

- **One unit, not two clocks.** A ratio between two *independent* clocks is an observable: it could come out differently. Here both terms are counted in one unit, and seventeen is fixed by the length of the cognitive process. The ratio is a definition that makes the system consistent, not a finding.
- **Deterministic against statistical.** In a causal set, chain length tracks proper time only on average, over a random sprinkling. The Abhidhamma's count is exact.
- **No chain is selected, on the usual reading.** A causal set's proper time belongs to the *longest* chain between two events, whichever it is. A stream is one moment at a time, so between two of its moments the longest chain is the only one, and the *Aṅguttara* sub-commentary gives every mind the same moment (*sabbassapi cittassa samānakhaṇattā*, `s0401t.tik.xml:1666`). The *Milindapañha* grades seven kinds of mind, from those that "arise heavily and go on slowly" (*garukaṃ uppajjati dandhaṃ pavattati*) to a Buddha's, quicker "by an incalculable degree" (`s0518m.nrf.xml:2842` – `:2870`), without saying whether the moment or the process is quicker; the *Sāratthadīpanī*

reads a Buddha's speed as a brief dwelling in the life-continuum (*bhavaṅgaparivāso lahuko*, `vin01t2.tik.xml:341`), not a shorter moment. On those readings "longest" selects nothing; read as moment lengths, the Milinda's grades would narrow this break.

- **Order and time.** A count of steps along a given succession is what every discrete clock does. The causal-set-specific part of "time as chain length" is that there is no time apart from the order. Whether a stream's succession is its time in that sense is left open in §3.4, so this feature is neither claimed nor denied here.
- **The count itself is contested.** Seventeen is disputed by the *Vibhaṅga Mūlaṭīkā* and its *Anuṭīkā*, and fifty-one depends on the presence sub-moment, which the *Mūlaṭīkā* questions and the *Vibhāvinī* defends (§5.2).
- **Time dilation: a collision with a reading the texts do not state.** Rossi and Hall (1941) found the lower-momentum group of cosmic-ray mesotrons (muons) decaying about three times faster than the higher, measuring at Denver and at Echo Lake, Colorado; Bailey et al. (1977) found muons stored in the CERN ring at a Lorentz factor of about 29.3 living about twenty-nine times their lifetime at rest. So a **per-kalāpa count taken as the same in every frame**, whether as each perceiver's count or as one Newtonian unit, would conflict with universal dilation. The texts state neither: the manual counts an object's age in the moments of the process that perceives it (ch. 4 §10, `abh07t.nrf.xml:1237`), but the *Vibhāvinī* glosses that count as a unit (`abh07t.nrf.xml:4849`, §5.3). A muon, moreover, is a series (*santati*), not a kalāpa, and the texts let a series' duration vary: "as series, phenomena are short-lived or long-lived as opposing or favourable conditions meet; by their own characteristic they are of one measure" (*pabandhavasena hi … parittakālā dīghakālā ca honti, sabhāvalakkhaṇavasena pana ekaparicchedā*, *Dhammasaṅgaṇī Anuṭīkā*, `abh04t.nrf.xml:821`). They also make an interval only within its own series (`abh03t.tik.xml:4309`). So the measured muon meets no textual claim directly: what conflicts with dilation is an unmeasured, frame-independent per-kalāpa count; the place of a duration within its series is textual; physics adds the Lorentz factor. The verdict is a **collision with a reading**, never *agreement*.
- **An upper bound on the sub-moment, conditional and ours.** The texts give the rate of mind only as floors (§5.2), and a measurement can supply only a ceiling, under a supposition no text makes. Suppose a Z boson were a series of pure octads. Production of matter by temperature happens at the producer's presence phase (*ṭhitippattāva*, manual ch. 6, `abh07t.nrf.xml:2213`), so the earliest change a measurement could see comes one sub-moment after the series begins. The Z's measured width, 2.4952 ± 0.0023 GeV (ALEPH et al. 2006), gives $\hbar/\Gamma \approx 2.64 \times 10^{-25}$ s, so **one sub-moment is at or below about 2.6×10^{-25} s**, and a mind-moment at or below about 8×10^{-25} s (`d4_bound_is_one_submoment`). A bound fifty-one times stronger, on the whole life of the kalāpa, was tried in the source run and does not follow from those premises (`d4_the_stronger_bound_does_not_follow`). With the tenth-of-a-second snap of §5.2, this is more than 10^{23} mind-moments per snap, above every textual floor and in conflict with none. There is no floor, no lattice and nothing to measure on the text's side. The verdict is an extension of the reading: philosophy, and no agreement with anything.
- **No physics prediction.** A unitless ratio fixed by consistency cannot predict a physical measurement. The only empirical programme it could seed is in perception science (calibrate a mind-moment against one measurement, predict an independent one). That is a separate programme with its own pre-registration, and not a physics claim.

5.5 · What would falsify this row

- **A text stating the lifetime of matter in units independent of the mind-moment**, such as a count of material moments or an external measure. The "one unit" reading would fail, and the ratio would become an observable.
- **A text that counts the mindless beings' matter, or matter outside every being, in some present mind's moments** (a perceiver's, say). The two cases of §5.3 would collapse into one, and a per-perceiver count, with its collision with time dilation (§5.4), would become the texts' own.
- **A text that counts the aeon in mind-moments**, or relates the two reckonings otherwise than by the divergence of §5.3. The aeon might then supply a second chain for the ratio.
- **A reading on which seventeen is observed rather than derived.** For example, a commentarial passage might present seventeen as reported meditative experience, independent of the cognitive-process arithmetic. The "consistency condition" reading would then weaken.

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6 · The Row We Dropped, and Why

The first version of this lens had a fifth row: *saṅkhāra*, the conditioned as a whole (*sabbe saṅkhārā*, "all conditioned things"), against Fotini Markopoulou's **quantum causal histories**. That framework puts quantum state spaces on a causal set, and so joins causal order to quantum structure.

It was cut before drafting, for two reasons.

1. **Category error.** *Saṅkhāra* includes the mental (*nāma*): consciousness, feeling, perception, volition. Nothing in a spin network or a quantum causal history models mind. A row that maps a category containing mind onto a structure that contains none is not a correspondence with a break. It is a correspondence that breaks everywhere.
2. **No structure added.** Every structural feature the row could claim was already in rows 1–4: order in row 1, grains and adjacency in rows 2–3. The row said that the conditioned as a whole resembles order-plus-quanta as a whole. A resemblance with no new structure is decoration: a coherence that feels like evidence and is none.

The neighbour, stated accurately. Markopoulou's 2000 paper assigns a finite-dimensional Hilbert space (a quantum state space) to each **event** and evolution operators between sets of events. The same paper already notes a problem: the evolution does not see the individual causal relations. It proposes, as an alternative, attaching the Hilbert spaces to the **edges** (the covering relations) and the operators to the events. On that reading, the state space on an edge from p to q is "the state space of p as seen by q". Hawkins, Markopoulou and Sahlmann (2003) then revised the definition. They put operator algebras on the events and made **each causal relation** carry a completely positive map, the most general quantum transformation, so that the evolution respects every individual relation.

There is an irony here, and it is recorded rather than used. Both moves make the **relation** a carrier of structure. That fits the *Paṭṭhāna*'s relation-first typing better than the original event-first definition did, but it does not rescue the row. A typed relation carrying a completely positive map is still not a relation of *object*, *predominance* or *kamma*.

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7 · Why a Theravāda Space, and Not Nāgārjuna's

A widely read recent meeting of Buddhist thought and physics is Carlo Rovelli's reading of **Nāgārjuna** (cross-tradition: Madhyamaka) beside relational quantum mechanics, in *Helgoland*. On the relational view, a system has properties only relative to another system. On Nāgārjuna's, nothing has own-being (*svabhāva*): everything is empty of intrinsic existence. Rovelli finds the two congenial.

This paper does not use that pairing, for a reason of structure and not of loyalty. **Madhyamaka denies own-being (*svabhāva*) to every dharma**, so it admits no ultimately real units to put in adjacency. **The Theravāda Abhidhamma is realist about dharmas**: its moments, its material phenomena and its conditions are real events. A discrete theory needs discrete units and relations among them. The Theravāda supplies both, so of the two it is the framework that *can* be compared structurally with causal sets and spin networks. It is also the framework whose space can be put beside a link: space as the delimitation between real units. One point weakens the contrast and should be stated. The boundary at the center of the adjacency row is defined as lacking own-nature: it is one of the ten kinds of matter the manual classes as not concretely produced (*anipphanna*; delimitation, the two intimations, the three mutabilities and the four characteristics, `abh07t.nrf.xml:2117` – `:2121`). The units are real (*nipphanna*, concretely produced); their delimitation is not a unit. So the argument for the Theravāda over Madhyamaka rests on the **units**, not on the boundary: a discrete comparison needs real units, and of these two frameworks only the Theravāda supplies them. It is not the only Buddhist school that does; the Sarvāstivāda of §4.4, also realist about dharmas, supplies them too. The Madhyamaka comparison is a comparison of attitudes toward relation. It is real, but it is a different comparison.

The institution's ruling, stated plainly. The institution publishing this paper has ruled, for its own purposes, that **delimiting space (*paricchedākāsa*) is conditioned and only nibbāna is unconditioned**, and that the place named *space* on its own organizational map of four bodies (§11) is delimiting space: never open space, and never the unconditioned. The ruling follows the manual and its sub-commentaries in reading the space element as delimiting matter, and it declines the Sarvāstivāda reading of *ākāśa* as unconditioned. Its ground is the boundary relation. On that map, space is the between of the four and exists only as their boundary, which makes it delimiting space whatever the texts below say of open space; in the texts' own terms, delimiting space has the matter it delimits as its proximate cause (§4.3).

The texts bear on the ruling at three points, and at the third they disagree.

1. The canon's *Kathāvatthu* refutes "space is unconditioned" (VI.6, §460–461, `abh03m3.mul.xml:7325` – `:7345`).
2. At the close of the same debate (§462, `abh03m3.mul.xml:7349`) the Theravādin also declines to call space *conditioned* without qualification. The *Kathāvatthu* commentary treats §§460–462 in a single paragraph, says that the rest of the debate is plain in meaning (*ṡesamettha uttānatthamevā*), and does not comment on that refusal (`abh03a.att.xml:4145`). Reading the refusal through the commentary's three spaces, as an earlier version of this section did, is this paper's reading and not the commentary's.
3. That paragraph names three spaces. It classes delimiting space as conditioned, and the other two, the space left by removing a meditation disk (*kasiṇa*) and unentangled, open space, as mere concepts (*paññattimattā*). For open space the classification is disputed. The *Atthasālinī* reads the *Dhammasaṅgaṇī*'s space element, "untouched by the four great elements" (§637, `abh01m.mul.xml:5473`), as unentangled space (*nijjaṭākāsaṃva kathitaṃ*, `abh01a.att.xml:8465`). The *Vibhaṅga Mūlaṭīkā* reads the *Sammohavinodanī*'s gloss on the

same words in the *Vibhaṅga* as speaking of unentangled space and not of the space left by the disk (`abh02t.tik.xml:653`, on `abh02a.att.xml:1093`). On that reading open space is the space element itself, which the *Dhammasaṅgaṇī* lists among material phenomena (*rūpaṃ ākāsadhātu*), and not a concept. The sub-commentaries on the *Dhammasaṅgaṇī* and on the manual gloss the same words as delimitation (`abh01t.tik.xml:2177`; `abh07t.nrf.xml:5821`). Ledi Sayadaw sets the commentary's reading beside a sub-commentary's, and his own gloss of the words is the delimiting one (`e0301n.nrf.xml:4077`).

So "open space is a mere concept" is the *Kathāvatthu* commentary's reading. The ruling keeps it as a second support, labeled as one of two readings, and its ground, the boundary relation, does not depend on it. On either reading, as this paper reads them, no space is left unconditioned: a concept is not an unconditioned phenomenon, and the *Dhammasaṅgaṇī* calls all matter conditioned (*sabbaṃ rūpaṃ … saṅkhatameva*, §594, `abh01m.mul.xml:5285`). The text nearest the other option is the *Milindapañha*, in which Nāgasena names space and nibbāna as the two things in the world born neither of kamma, nor of a cause, nor of temperature (`s0518m.nrf.xml:5644`), without calling either unconditioned in that passage. The Sarvāstivāda draws a comparable line: beside its unconditioned *ākāśa* (*Kośa* I.5), it names a conditioned space element, the cavity (I.28), as §4.4 records.

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8 · The Strongest Objection, Carried

Objection. "The Abhidhamma is a phenomenology: an analysis of what appears in experience, made for the purpose of liberation. Quantum gravity is a theory of what spacetime is. The kalāpa contains colour, odour and taste, which are qualities of matter *as sensed*. The *Kathāvatthu* commentary says that space between two trees is known by the **mind**, from the absence of visible matter, not seen by the eye. A comparison of the two frameworks compares an account of knowing with an account of being. Any shared shape is a coincidence of two different questions."

What survives it. Most of the objection stands, and the paper does not answer it away. Every row compares **formal shape**: an acyclic relation, a boundary located at an adjacency, a duration expressed as a count. Shape is what two frameworks with different subject matter can share without sharing a subject, and it is all this paper claims they share. The objection does not show that the shapes differ. It shows that shared shape proves nothing about shared subject matter, and the paper agrees. What the objection *does* remove is any reading of the lens as evidence that the Abhidhamma knew physics, or that physics vindicates the Abhidhamma. Neither reading was offered, and the objection makes sure neither can be taken.

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9 · The Counter-Argument to This Paper's Strongest Claim

The strongest claim is the adjacency row (§4.5). Against it: **the idea that a boundary lives at the adjacency is not specific to loop quantum gravity.** It holds in any cellular picture of extended stuff: a cell complex, a Voronoi tessellation, a chessboard, and Regge calculus (Regge 1961), in which curved spacetime is approximated by flat simplices glued along shared faces, and the geometry is fixed by the edge lengths of neighbouring cells. The Western literature on boundaries has argued the ownership question directly. When two bodies touch, does the boundary belong to one of them (Bolzano), to neither (a view the literature traces to Leonardo), or is it two coincident boundaries, one belonging to each (Brentano)? (Varzi's survey; Casati and Varzi 1999, ch. 5.) The "neither" view is the nearest Western neighbour of the manual's "a constituent of neither". Brentano's is a contrast: his coincident boundaries are dependent parts of the bodies they bound, which is the opposite of the *Vibhāvinī*'s delimiter, untouched by both. Aristotle defined place as the innermost boundary of the containing body (*Physics* IV.4), a boundary account of *where* that is two millennia old and owes nothing to quanta. The *Kośa*'s debate (§4.4) shows the Indian schools arguing the same question in the first millennium. If the shared shape is generic, then the match with loop quantum gravity specifically is weaker than the table makes it look.

What loop quantum gravity adds to the generic picture is that the adjacency carries a **quantised magnitude**, and that volume is **derived** from it. Those are exactly the two features the Abhidhamma lacks (§4.6, breaks 1 and 2). So the part of the row that is loop-quantum-gravity-specific is the part that fails, and the part that holds is generic. The row survives only in this form: *the Theravāda sub-commentary's answer to "what is between minimal units?" has the shape of the answer loop quantum gravity gives, and not the shape of the answer an earlier Indian school gave.* The paper does not refute the generic reading. It claims no more than that.

The same counter-argument, run on the ratio row (§5.3). Against it: every discrete clock counts steps, so "a duration is a count" is not specific to causal sets. The causal-set-specific features are three: proper time is the length of the **longest** chain, whichever chain that is; the relation between chain length and proper time is **statistical**, holding on average over a random sprinkling; and there is **no time apart from the order**. The first two are among §5.4's breaks, and the third is left open (§3.4). So, as with the adjacency row, what holds for certain is generic: a duration stated as a count of steps (§5.3).

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10 · What Would Falsify These Findings — Collected

Row	The finding	Falsified by
1 Order	the earlier→later sub-relation S, over condition-modes, is acyclic (by selection); a strict order, not yet a causal set, can be read over it; S also orders moments of different streams; the rest is extra typing	a member of S with the conditioned arising at or before the condition, in any layer; cross-moment kamma with a present or future relatum; an implied temporal order S cannot connect (engaged in a narrowed form between streams, §3.4; open within a stream)
2–3 Adjacency	both put the unit of spatial boundary at the adjacency between minimal units, and neither makes it a unit; the shared part is generic; the between is matter's and never mind's, for the tradition's own reason	a text making delimiting matter a kalāpa constituent, or placing space inside a cluster; a text with two moments of one stream present at once, with mind given shape, or with space delimiting mental phenomena; a formulation of loop quantum gravity in which area between adjacent grains is not carried by their adjacency; a reading of the *Vibhāvinī*'s space as mere absence (partly engaged, §4.7)
4 Ratio	matter dated on a stream's own moments where there is one, a unit where there is none; only the generic part survives, a duration as a count of steps; the aeon meets the mind-moment only as divergence	a text timing matter independently of mind-moments; a text counting the mindless beings' matter, or external matter, in some present mind's moments; a text counting the aeon in mind-moments; seventeen shown to be observed rather than derived
The scale (§5.2)	the snap is named, not timed; the rate is round lower bounds in four works, with no ceiling	a text giving the finger-snap a duration, or the rate an upper bound
The status of continuity (§2)	relative length, number, shape and age are designations for mind and matter on the teachers' reported method; the mind-only rows constrain no matter; *santati* of the mental stream is not a continuum	a canonical or commentarial text making length or number real by own-nature; a mind-only row shown to entail a material claim; a text giving mental moments no immediate successor
The grain's size (§4.2, §4.8)	the count follows from time, and the size does not; the *Mahāṭīkā*'s size, read metrically and carried outward, collides by count and by size; Ledi's bound escapes, leaving no grain of extent	a text sizing a kalāpa of external matter; a text making the *Mahāṭīkā*'s line a measurement; a reading on which a nucleus holds no plurality of dhammas; the *Mahāṭīkā*'s reading found to bear a sense Ledi's objection does not touch
The sub-moment bound (§5.4)	at or below about 2.6×10^{-25} s, if the Z is a series (ours)	a corrected earliest-change argument under A64; a text making the sub-moment a fixed duration; the Z's width revised
The dropped row	*saṅkhāra* ↔ quantum causal histories adds no structure	a structural feature of quantum causal histories with a *Paṭṭhāna* counterpart not already in rows 1–4

11 · Stake, and What Was and Was Not Done

The authors' stake. The authors write from within the Theravāda. The institution publishing this paper grounds its values in the Pāli canon, has ruled the Theravāda reading of space over the Sarvāstivāda one, and uses the word *space* on an organisational map of its own. Nothing in this paper bears on that map, and the ruling was made so that the map would not borrow a predicate it should not have. There is a pull, for authors in this position, to see the tradition anticipate frontier physics; the founder's own question began as a hope that the Abhidhamma might hold the bridge between the discrete and the smooth. **The pull was acted on, and caught.** An adversarial review before any prose dropped a row, merged two and narrowed the order row; outside review caught summaries that outran their own sections; and the 2026-09-30 revision withdrew three statements of the deposited text: that matter is "timed by the mind's chain" (§5.3), that nothing in S orders moments of different streams (§3.4), and that the finger-snap figure is one rhetorical number (§5.2). The 2026-10-06 revision withdrew a fourth: that the space ruling of §7 follows the texts at each point, when the commentary's classification of open space is disputed within the tradition. The 2026-11-03 revision withdrew a fifth: that the kalāpa's identification with the *paramāṇu* is "later exegesis, common in modern summaries" (§4.2), when a sub-commentary makes it and Ledi Sayadaw declines it. Its source run was scored on whether it would report the collision of §4.8; that collision was foreseeable once the question named the size, so the run tests reporting and not discovery, and its refuter killed two of its first statements ("three voices", which are two authors, and a collision "on every textual size", which Ledi's bound escapes) and cut a bound by fifty-one times. The sibling paper's first framing made the same kind of error in the other direction. The reversals are part of the result.

Done:

- The *Dhammasaṅgaṇī* (§637), the *Vibhaṅga* (space element), the *Kathāvatthu* (VI.6, §460–462; VI.7, §463–464) and its commentary, the *Visuddhimagga* (ch. XIV, ¶442; the kalāpa passages), the *Sammohavinodanī* (the size ladder; the finger-snap), the *Abhidhammatthasaṅgaha* (chs. 3, 4, 6, 8) and the *Abhidhammatthavibhāvinī-ṭīkā* (on chs. 3, 4, 6, 8), all read in the Chaṭṭha Saṅgāyana edition;
- the *Abhidharmakośabhāṣya* ad I.43 read in the GRETIL Sanskrit (after Pradhan);
- Rovelli's Zakopane lectures, the Rovelli–Upadhya primer, Markopoulou (2000) and Hawkins–Markopoulou–Sahlmann (2003) read in full; Sorkin (2007) and Dowker (2022/2023) read in full for the sibling paper;
- the census run as described;
- in the 2026-09-28 revision: Karunadasa (1967, chs. 5 and 7) and Bodhi (1993, the introduction and chs. VI and VIII) read for the passages cited; U Nārada (1969) consulted only for its treatment of the conditions' modes; the *Abhidharmakośabhāṣya* ad I.28 and II.22 read in the GRETIL Sanskrit; the *Paṭṭhāna* searched as described under *Not done*;
- in the 2026-09-30 revision: every Pāli locus given with a file and line read at VRI `tipitaka-xml` commit `05d5d3c`, with fifteen lines of context and a mechanical check of each quotation; `kala/Time.lean` and `kala/Axioms.lean` (commit `deb9dec`) read and their run script re-run (seven theorems check, six breaks fail); the run *rate-of-mind control* read for its three floors only; Rossi and Hall (1941) read in a scanned copy; and the revision's draft put through an adversarial review, whose corrections it carries;
- in the 2026-10-06 revision: the loci of §7 (the *Kathāvatthu* and its commentary, the *Dhammasaṅgaṇī* and the *Atthasālinī*, the *Sammohavinodanī* and the *Vibhaṅga Mūlaṭīkā*, the *Dhammasaṅgaṇī Mūlaṭīkā*, the *Vibhāvinī*, Ledi Sayadaw's *Paramatthadīpanī* and the *Milindapañha*) read at the same commit with fifteen lines of context; every other Pāli locus checked again against its quotation or the statement it supports; and the physics and secondary citations checked against publisher, archive or index records, as listed under *Sources checked at the 2026-10-06 revision*;
- in the 2026-11-03 revision: every new Pāli locus of §2, §4.2 and §4.8 read at the same commit; the formal-abhidhamma run *the continuum bridge* (PREREG, DERIVATION, REFUTER, RUN and `Bridge.lean`, commit `a964ee1`) read in full; Crossref records opened for Rutherford (1911), Hofstadter (1956), Pohl et al. (2010) and ALEPH et al. (2006). Rutherford's 3.4×10^{-14} m was read by the run's refuter in a transcription; the width of the Z was fetched in that run.

Not done:

- **The *Paṭṭhāna* itself was not read in full.** The conditions are characterised here through the manual and the sub-commentary. In this revision the Chaṭṭha Saṅgāyana *Paṭṭhāna* was searched for specific passages only: kamma across moments (*Kusalattika* §427), and the modes of presence, dissociation and support. A canonical-layer check of row 1, including whether the *Paṭṭhāna* anywhere composes two conditions, is owed; U Nārada's translation (1969) is the route.
- The *Mūlaṭīkā*s on the *Vibhaṅga* and the *Paṭṭhāna*, their *Anuṭīkā*s and Ledi Sayadaw's two works were read only at the passages cited, not in full.
- Bailey et al. (1977) was not reached (a login page). Its figures are checked against a re-analysis of its data (Lobo, Pfeifer and Morais 2025: a momentum of 3.094 GeV/c, and a dilated lifetime of 64.378 ± 0.026 microseconds against 2.197 at rest, so a Lorentz factor of about 29.3).
- No text was found that counts matter without a mind in mind-moments. None of the 35 sub-commentary passages on the mindless beings that the reading tool finds does, and the rest of the tradition was not searched for it; the absence is reported for that aperture only.
- Von Rospatt (1995) was not reached, and Casati and Varzi (1999) was cited through Varzi's survey, not read.
- Apart from Ledi Sayadaw's passages, the later and modern literature on kalāpas, in Burmese, Thai and Sinhala, was not read.
- Vasubandhu's *Viṃśikā* critique of partless atoms, and Aristotle's *Physics* IV, were cited from standard knowledge and not re-read for this paper.
- Apart from the arithmetic of §4.8 and §5.4's bound, no physics was calculated, and no physicist has reviewed the primers or those two passages. The primers were checked against the cited sources only. The gold nucleus's size and nucleon count, the proton's and the sunbeam threshold are stated from standard knowledge or a rough figure, not from a measurement read for this paper.
- The census was quick and English-only. The full census has not been run.

12 · Honest Limits

- **This is a lens.** No row predicts a measurement, and no row can. The comparison is of formal shapes, and shared shape is not evidence of shared subject matter (§8).

- **Two of the four rows are weak by their own account.** The grain row survives only as a presupposition of the boundary row. The ratio row is a definition, not a finding, and its one contact with measurement, time dilation, is a collision with a reading (a frame-independent per-kalāpa count) that the texts do not state.
- **The strongest row lives at a late layer.** The adjacency correspondence holds for the sub-commentary (c. 12th century), not for the canon, whose space is a cavity.
- **The physics is secondary, and simplified.** The primers state standard results and cite their sources. They omit the dynamics of both theories, which are unsettled, and the area formula is given for transversal crossings only.
- **Rows 2 and 3 presuppose three things the physics does not settle.** The area spectrum used is the **kinematical** one; whether it survives the imposition of the constraints is disputed. Definite adjacency holds only for a **spin-network basis state on a fixed graph**; a general state superposes graphs. And the grains-and-faces picture is the **semiclassical** polyhedral reading, least trustworthy at the small spins where the smallest area sits.
- **Row 1 is not a causal set,** for the four reasons of §3.4, and its order between streams is the reader's composition of edges the texts state separately.
- **The machine check is small.** The Lean theorems of §4.5 check the within-stream precondition only, in an exploratory model the authors wrote; adjacency and shape are not modelled.
- **The two cases of §5.3 rest partly on an absence.** No text found counts matter without a mind in mind-moments; a text that did would overturn the second case (§5.5).
- **Two questions are left open.** Whether a stream's succession is its time in the causal-set sense (§3.4), and which reading of §5.3 the seventeen bears where no mind is present.
- **§4.8 is a collision with a stack of this paper's own steps.** The anchor, the metrology, the outward carrying and the premise that what is individuated in a kalāpa must be a member are ours, and any one could be refused. A nucleon is not a dhamma in any text, so a reader may hold that a nucleus's count and a kalāpa's count do not meet at all. The collision bears on the reading it names and on nothing else in the tradition.
- **§5.4's bound rests on a supposition no text makes.** A Z boson is not a *santati* in any text, and treating a width as a delay is the simplest reading of a quantum resonance, not a derivation.
- **The continuity finding of §2 is partly true by construction.** That mind-only rows constrain no matter follows from a model in which they have no symbol for it, and the model and its row list are the authors'.
- **The institution's space ruling is its own.** Its ground is the boundary relation on the institution's map; the classification of open space as a concept is one of two readings of a disputed point (§7). No finding of this paper depends on either.
- **The rate figures are floors, and their conversion to seconds is the authors'.** The finger-snap's tenth of a second, and the reading of the figures as round orders of magnitude, are assumptions of this paper, not of the texts.
- **Translations are the authors'.** Where a Pāli or Sanskrit term has a contested rendering (*anipphanna*, *kālavimutta*, *ākāsakoṭṭhāsika*), the rendering chosen is a reading, not a settled equivalence.
- **Edition.** Paragraph numbers follow the Chaṭṭha Saṅgāyana edition. The Pali Text Society's numbering differs; passages should be located by title and quotation.
- **The census aperture was narrow.** Buddhism-and-physics comparison is a large, poorly indexed literature, much of it in books and in languages not searched. Absence from the search is weak evidence.

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13 · Close

The *Kathāvatthu* commentary gives a small example. A person looks at two trees. The eye sees the trees. The mind, finding no matter between them, knows "space". Loop quantum gravity, asked where the space is between two grains, also answers *between*, and puts a number on it, which the text does not. The shared part is the generic one (§9). The two answers share that shape and have two subjects. This paper records the shape and marks the boundary between the subjects, which, as in the text it reads, belongs to neither side.

Cross-Venue References

Venue	Identifier
Primary canonical	https://thonly.org/research/abhidhamma-and-discrete-quantum-gravity
GitHub	https://github.com/thonly/publications/blob/main/defensive-publications/abhidhamma-and-discrete-quantum-gravity.md
Sibling paper	https://thonly.org/research/the-vibhajjavadin-view-of-time
Census pre-registration (public)	`thonly/publications` commit `7348ba2`, `timestamps/census/physics-lens-prereg.md`
Sibling paper (the Paṭṭhāna's relations)	https://thonly.org/research/patthana-typed-causation-vocabulary
Lean 4 formalisation (public, CC0)	https://github.com/SiliconWat/formal-abhidhamma (`kala/Time.lean`, `kala/Axioms.lean`; `runs/2026-10-14-the-continuum-bridge/Bridge.lean`); software concept DOI 10.5281/zenodo.23067241

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25. Lobo, I.P., Pfeifer, C. & Morais, P.H. (2025). "Experimental Bounds on Deformed Muon Lifetime Dilation." *Physics Letters B* 866: 139511. arXiv:2406.05150v3 (re-analyses the data of 24).

Pāli and Sanskrit sources (Chaṭṭha Saṅgāyana edition, VRI `tipitaka-xml`, unless noted; file:line references are to the `romn/` text at commit `05d5d3c`)

26. *Dhammasaṅgaṇī* §637, the space element (`abh01m.mul.xml:5473`); §594, all matter conditioned (`abh01m.mul.xml:5285`); §652, kamma-produced visible form (`abh01m.mul.xml:5549`).
27. *Vibhaṅga*, *Dhātuvibhaṅga*, the space element; and the modes of existence, including non-percipient existence (`abh02m.mul.xml:3685`).
28. *Dhātukathā*, non-percipient existence as one aggregate (`abh03m1.mul.xml:421`).
29. *Kathāvatthu* III.11 *Asaññakathā* (`abh03m3.mul.xml:5797`); VI.6 *Ākāsakathā*, §460–462 (`abh03m3.mul.xml:7325`–`:7349`); VI.7, §463 (`abh03m3.mul`).
30. *Kathāvatthu-aṭṭhakathā* on VI.6–7: the three spaces (`abh03a.att.xml:4145`); the two trees (`abh03a.att.xml:4157`).
31. *Paṭṭhāna-aṭṭhakathā*: on contiguity and the absence of shape (`abh03a.att.xml:7697`); on the root condition, kamma-born matter going on without mind, and non-percipient existence as the place of matter (`:7857`, `:7861`).
32. Buddhaghosa. *Visuddhimagga* ch. XIV, ¶442 (the space element), the kalāpa passages of the second volume, and the death-and-rebirth passage (`e0101n.mul`, `e0102n.mul`; `e0102n.mul.xml:4045`).
33. *Sammohavinodanī* (*Vibhaṅga-aṭṭhakathā*): the size ladder (gloss on *aṇumattesu vajjesu*); the finger-snap figure for feeling (`abh02a.att.xml:533`); the lifetime of matter, the two walkers, and the older and replacement rules for dating kamma-born matter (`abh02a.att.xml:409`–`:433`); the death-and-rebirth passage (`abh02a.att.xml:2889`); the external space element (`abh02a.att.xml:1093`).
34. *Atthasālinī* (*Dhammasaṅgaṇī-aṭṭhakathā*), on internal and external, on time as a designation, and on the space element as unentangled space (`abh01a.att.xml:3485`, `:3721`, `:8465`); and the *Dhammasaṅgaṇī Mūlaṭīkā* on §637 (`abh01t.tik.xml:2177`).
35. *Vibhaṅga Mūlaṭīkā* and *Vibhaṅga Anuṭīkā*, on the lifetime of matter, the presence sub-moment and the space element (`abh02t.tik.xml:305`, `:309`, `:313`, `:653`, `:4933`, `:4937`).
36. *Paṭṭhāna Mūlaṭīkā* and *Paṭṭhāna Anuṭīkā*, on contiguity, shapelessness and the time-interval across the mindless existence (`abh03t.tik.xml:4305`, `:4309`; `abh05t.nrf.xml:4637`, `:4641`).
37. Anuruddha. *Abhidhammatthasaṅgaha* chs. 2, 3, 4, 6, 8 (`abh07t`, bound with the sub-commentary): among others `abh07t.nrf.xml:329`, `:1229`–`:1241`, `:1485`, `:1693`, `:1957`, `:2117`–`:2121`, `:2169`, `:2213`, `:2273`, `:2277`, `:2349`, `:2865`, `:2873`, `:2909`, `:2913`.
38. Sumaṅgala. *Abhidhammatthavibhāvinī-ṭīkā* on chs. 3, 4, 6, 8 (`abh07t`): among others `abh07t.nrf.xml:4833`–`:4849`, `:5553`, `:5821`, `:6653`.
39. Vasubandhu. *Abhidharmakośabhāṣya* ad I.5, I.28, I.43 and II.22, ed. P. Pradhan (2nd ed., 1975), pp. 32–33 for I.43; GRETIL e-text (line references in square brackets are GRETIL's page|line).

40. *Paṭṭhāna*, *Kusalattika*, *pañhāvāra* (`abh03m7.mul`): §427 (kamma across moments, `abh03m7.mul.xml:6065`); the modes of the presence, dissociation and support conditions.
41. *Brahmajāla Sutta* (DN 1), the non-percipient beings (`s0101m.mul.xml:487`).
42. *Saṃyutta Nikāya* 15.5 (*Pabbata*) and 15.8 (*Gaṅgā*), on the length and number of aeons (`s0302m.mul.xml:3529` – `:3533`, `:3565`).
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44. *Sāratthappakāsinī* (*Saṃyutta* commentary), on SN 12.61, the rate of mind per finger-snap (`s0302a.att.xml:1463`, `:1467`).
45. *Manorathapūraṇī* (*Aṅguttara* commentary), the finger-snap as the striking of two fingers (`s0401a.att.xml:1119`); and the *Aṅguttara* sub-commentary on AN 1.48, every mind of the same moment (`s0401t.tik.xml:1666`).
46. *Paramatthajotikā* (*Suttanipāta* commentary), two minds never together (`s0505a.att.xml:685`).
47. *Sāratthadīpanī-ṭīkā* (Vinaya sub-commentary), the rate of mind and the discerned and undiscerned orders (`vin01t2.tik.xml:341`).
48. *Dhammasaṅgaṇī Anuṭīkā*, the rate of mind, and the durations of series (`abh04t.nrf.xml:821`).
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70. Sub-commentary on the *Abhidhammāvatāra*, on the eye-sensitivity, `abh08t.nrf.xml:7373`.

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Sources checked at the 2026-10-06 revision

Each record below was opened on 2026-10-06 before the work was cited or kept. Where a record lacked a detail the text uses, the detail was checked against a search index's summary that day and is marked so.

- Bombelli, Lee, Meyer & Sorkin (1987): INSPIRE-HEP record (pages 521–524); the publisher's page refused automated access.
- Brightwell & Gregory (1991): Crossref record for doi:10.1103/PhysRevLett.66.260 (title, volume, issue, pages). The result attributed to it in §3.1 was not re-read.
- Rideout & Sorkin (2000); Markopoulou & Smolin (1997); Bianchi, Doná & Speziale (2011); Hawkins, Markopoulou & Sahlmann (2003): arXiv records gr-qc/9904062, gr-qc/9702025, 1009.3402, hep-th/0302111 (journal references; the polyhedral reading and its semiclassical status from the Bianchi–Doná–Speziale abstract; matrix algebras on events and completely positive maps between causally related events from the Hawkins–Markopoulou–Sahlmann abstract).
- Sorkin (2007): arXiv gr-qc/0703098, full text. The remark on "accretive time" and the "Vibhajavadin" school is a footnote: "The notion of 'accretive time' that arises here seems close to that of C.D. Broad, and also to that of the 'Vibhajavadin' school within the Buddhist philosophical tradition." The paper's quotation elides the reference to Broad.
- Dowker (2022/2023): arXiv 2209.07653v2, full text. It quotes Sorkin's footnote, and its own first footnote proposes, as a project for a scholar of Buddhism interested in physics, to determine the "Vibhajavadin" view of time, perception and epistemology "by examining the earliest existing records".
- Wüthrich & Callender (2017): Crossref record for doi:10.1093/bjps/axv040.
- Rovelli & Smolin (1995): Crossref record (pages 593–619; the arXiv record gives 593–622, now corrected); erratum from the arXiv record.
- Ashtekar & Lewandowski (1997): Crossref record (pages A55–A81; the arXiv record gives A82, now corrected).
- Loll (1996): arXiv gr-qc/9511030, full text. It states that the three-valent spectrum is "identically zero", citing Loll (1995) for the result, and that states of valence below three are annihilated; §4.1's attribution is corrected accordingly. Loll (1995): arXiv gr-qc/9506014 (journal reference).
- Rovelli (2011), Zakopane lectures: arXiv 1102.3660, full text. The three quotations of §2 and §4.1 were matched word for word, in §I.A and §II.F.
- Markopoulou (2000): arXiv hep-th/9904009, full text. The problem (the evolution operator "does not distinguish the different possible underlying causal sets"), the alternative with Hilbert spaces on the edges and operators on the events, and "the state space of p as seen by q" (§5.3 and §7.2 of that paper) were matched.
- Domagala & Lewandowski (2004); Meissner (2004); Ghosh & Mitra (2005): arXiv records gr-qc/0407051, gr-qc/0407052, gr-qc/0411035 (journal references; Ghosh–Mitra pages 114–117). The values $\gamma \approx 0.2375$ and $\gamma \approx 0.274$ are not in the abstracts and were checked against a search-index summary of the literature.
- Hawking, King & McCarthy (1976); Malament (1977); Regge (1961); Rossi & Hall (1941); Bailey et al. (1977): Crossref records (titles, volumes, issues, pages).
- Rossi & Hall (1941): the scanned copy cited in item 23, read again: the less penetrating group decays "at a rate about three times faster than the more penetrating group", with measurements at Denver and at Echo Lake.
- Lobo, Pfeifer & Morais: arXiv 2406.05150v3, full text and record. The record gives publication in *Physics Letters B* 866 (2025) 139511, now added; the figures used in §11 (momentum 3.094 GeV/c, dilated lifetime 64.378 ± 0.026 μs, rest lifetime 2.197 μs) were matched.
- Cousins (2001): the publisher's article page and a search-index summary (volume 18, issue 2, pages 131–182).

- Varzi, "Boundary", *Stanford Encyclopedia of Philosophy*: read again; it assigns the boundary to one body or the other in Bolzano's account, to neither in Leonardo's, and to two coincident boundaries in Brentano's, and cites Casati and Varzi (1999), ch. 5.
- Ratnayaka (1991): bibliographic record from a search-index summary of the Pandit project's catalog; the article was not opened.
- *The Ephemeral Projector* (ResearchGate 397427972): the record refused automated access on 2026-10-06 and a search did not surface it; it is kept as found by the census on 2026-09-27.
- "Causal Set Theory and Abhidhamma: A Dialogue Between Modern Physics and Buddhist Philosophy", ekasiswanto.wordpress.com, 2026-07-31: the dated archive page, which carries the author's own note that the post is AI-generated.
- "Kalapa (atomism)": the Grokipedia article compares the kalāpa's scale to the Planck length; the Wikipedia article does not mention the Planck length, and gives the kalāpa the size of 1/46,656 of a chariot-dust mote, the *paramāṇu*'s figure (§4.2).
- The Pāli passages: every file-and-line reference in the paper was checked in the Chaṭṭha Saṅgāyana text at VRI `tipitaka-xml` commit `05d5d3c`, against its quotation, or, where none is quoted, against the statement it supports. The loci of §7 were read with fifteen lines of context: *Kathāvatthu* VI.6 (`abh03m3.mul.xml:7325` – `:7349`) and its commentary (`abh03a.att.xml:4145`); *Dhammasaṅgaṇī* §594 and §637; the *Atthasālinī* (`abh01a.att.xml:8465`); the *Sammohavinodanī* (`abh02a.att.xml:1093`) with the *Vibhaṅga Mūlaṭīkā* (`abh02t.tik.xml:653`); the *Dhammasaṅgaṇī Mūlaṭīkā* (`abh01t.tik.xml:2177`); the *Vibhāvinī* (`abh07t.nrf.xml:5821`); Ledi Sayadaw (`e0301n.nrf.xml:4077`); and the *Milindapañha* (`s0518m.nrf.xml:5644`, where Nāgasena speaks). Who speaks was checked for SN 15.5 (a monk asks, the Buddha answers), SN 15.8 (a brahmin asks, the Buddha answers) and DN 1. Two quotations end where the text adds the quotative *iti* (*purejātavasena*, *aññamaññaṃ*, manual ch. 8 §19 and §29); one quoted phrase, *aṭṭhakathāyaṃ āgatattā vuttaṃ*, stands at `abh02a.att.xml:429`, beside the rule at `:421` that it characterizes.
- Not re-checked at this revision: the *Abhidharmakośabhāṣya* (GRETIL and Pradhan); Karunadasa (1967), Bodhi (1993), U Nārada (1969) and von Rospatt (1995); Rovelli's *Helgoland*, Capra (1975), Aristotle and the *Viṃśikā*; Rovelli and Upadhya (1998), whose arXiv record was opened for its title only.